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PEE 511 — Past Questions & Materials
by Renance

Course: PEE 511
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 500

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Paper 1 of 2 — PEE 511/514

Level 500 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN,
DEPARTMENT OF PETROLEUM ENGINEERING**

COURSE TITLE: ENHANCE OIL RECOVERY
COURSE CODE: PEE 511
TIME: 3HRS
INSTRUCTION: ATTEMPT QUESTION NO. 1 AND ANY OTHER 3

2023/2024

Question One (1)

The following permeability data were measured on a sandstone as a function of its water saturation

S_w	0	10	20	30	40	50	60	70	75	80	90	100
K_{ro}	1.0	1.0	1.0	0.94	0.8	0.44	0.16	0.045	0	0	0	0
K_{rw}	0	0	0	0	0.04	0.11	0.2	0.3	0.36	0.44	0.68	1

Critical saturation or oil and water

- Plot the relative permeabilities to oil and water versus water saturation on Cartesian coordinate paper.
- Plot the relative permeabilities ratio versus water saturation on semi log paper.
- Find the constants a and b in the equation $K_o/K_w = ae^{-bsw}$ from the slope and intercept of your graph. Also find a and b by substituting two sets of data in the equation and solving simultaneous equations.
- If $\mu_o = 3.4\text{cp}$, $\mu_w = 0.684$, $\beta_o = 1.50\text{bbl/STB}$, and $\beta_w = 1.05\text{bbl/STB}$, what is the surface water cut of a well completed in the transition zone where the water saturation is 50%?
- What is the reservoir cut in part (d)?
- What percentage of recovery will be realized from this sandstone under high-pressure water drive from that from that portion of the reservoir above the transition zone which is invaded by water. The initial connate water saturation above water zone is 30%.
- If water drive occurs at a pressure below the saturation pressure such that the average gas saturation is 15% in the invaded portion. What percentage recovery will be realized. The average oil volume factor at the lowest pressure is 1.35bbl/STB and the initial oil volume factor is 1.50bbl/STB.
- What fraction of the absolute permeability of this sandstone is due to the least permeable pore channels that make up 20% of the pore volume? What fraction is due to the most permeable pore channels that make up 25% of the pore volume.

Question Two(2)

- Define the term EOR, show the target and path of EOR.
- Give any four (4) reasons why oil and gas companies embark in EOR activities.
- A saturated oil reservoir is under consideration to be water-flooded immediately after drilling and completion. Core analysis tests indicate that initial and residual oil saturations are 75 and 35% respectively. Calculate the displacement efficiency, E_D

when the oil saturation is reduced to 65, 60, 55, 50 AND 35%. Assume that the oil FVF, B_o remains constant throughout the project life.

Question Three (3) ✓

The following data in table 1 below were measured on a substance as a function of its water saturation:

Table 1.0: Relative permeability data of Irri field, in Isoko South Local Government

S_w	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.85	0.9	1
K_{ro}	1	0.99	0.93	0.62	0.35	0.2	0.12	0.06	0.015	0	0	0
K_{rw}	0	0	0	0.02	0.05	0.12	0.22	0.35	0.51	0.6	0.71	1

- Find the constants 'a' and 'b' by substitution in the equation

$$\frac{K_{ro}}{K_{rw}} = ae^{-bsw}$$
- If the formation contains oil of 2.2cp Viscosity, 0.75cp Viscosity of water, oil FVF of 1.65rb/stb and water FVF of 1.05rb/stb. What is the surface water cut of a well completed in the 53% transition zone?
- What is the reservoir cut in part (b) above
- List any four (4) forces affecting recovery.

Question Four (4) ✓

- Differentiate IOR and EOR and give four (4) examples of each
- Outline any four (4) practical considerations for every EOR designs
- Discuss the displacement of oil by water in stratified reservoirs
- Explain why the ultimate recovery factor is different for different oil recovery.
- What is a Fractional Flow of a Fluid?
- Any influence that causes the fraction flow curve to shift upward (i.e. increases in F_w or F_g) will result in a more efficient displacement process (TRUE or FALSE? Explain.

Question Five (5)

- Oil Recovery operations have been traditionally been divided into 3 stages. Explain these stages in a Chronological sense.
- Give at least two (2) examples of each stage of Oil recovery classification.
- Mention four (4) fluids used in EOR operations and their primary objectives.
- Where can you place gas lifting operation in the different recovery classification.

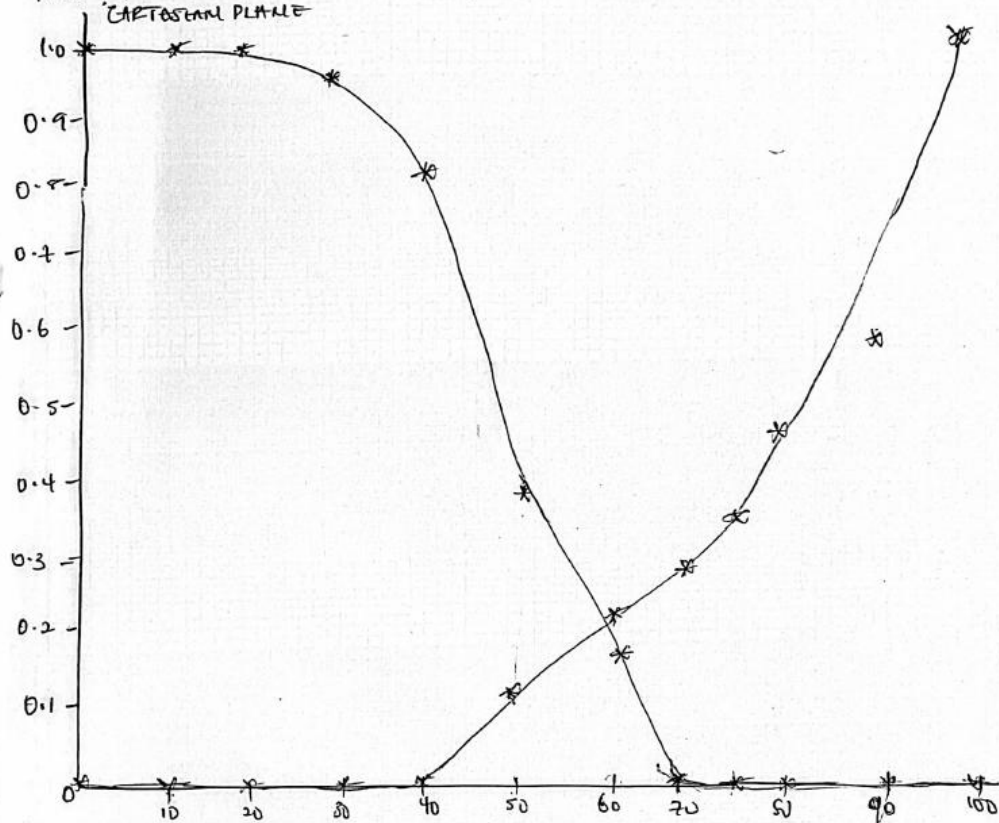
Examiner: Engr. Prof. P.E. Akpoturi.

SURE SUCCESS FOUNDATION

PEE 511 - ENHANCED OIL RECOVERY (EOR)
(2023/2024)

Question 1

Plot of relative permeabilities to oil and water Versus water saturation on Cartesian plane



Sw	0	10	20	30	40	50	60	70	75	80	90	100
kro	1.0	1.0	1.0	0.95	0.8	0.44	0.16	0.045	0	0	0	0
krw	0	0	0	0	0.04	0.16	0.2	0.3	0.36	0.44	0.64	1

$$4c) \frac{k_o}{k_w} = a e^{-b S_w}$$

$$\text{At } S_w = 40\% \quad \frac{k_o}{k_{rw}} = \frac{0.8}{0.04}$$

$$\frac{k_o}{k_{rw}} = 20$$

$$\text{At } S_w = 60\%, \quad \frac{k_o}{k_{rw}} = \frac{0.16}{0.2}$$

$$\frac{k_o}{k_{rw}} = 0.8$$

$$\frac{20}{0.8} = \frac{a e^{-b(0.4)}}{a e^{-b(0.6)}} \quad \text{--- (1)}$$

$$25 = e^{-0.4b + 0.6b}$$

$$25 = e^{+0.2b}$$

$$\ln 25 = \ln e^{+0.2b}$$

$$3.218 = 0.2b$$

$$b = \frac{3.218}{0.2} = 16.09$$

$$\therefore 20 = a e^{-b(0.4)} \quad \text{--- (2)}$$

To find a , substitute b in to equation 1

$$20 = a e^{-(16.09 \times 0.4)}$$

$$20 = a e^{-6.436}$$

$$a = \frac{20}{e^{-6.436}} = 12.478 \cdot 12$$

$$1d) \mu_o = 3.4 \text{ cp}$$

$$\beta_o = 1.50 \text{ bbl/stb}$$

$$\mu_w = 0.64 \text{ cp}$$

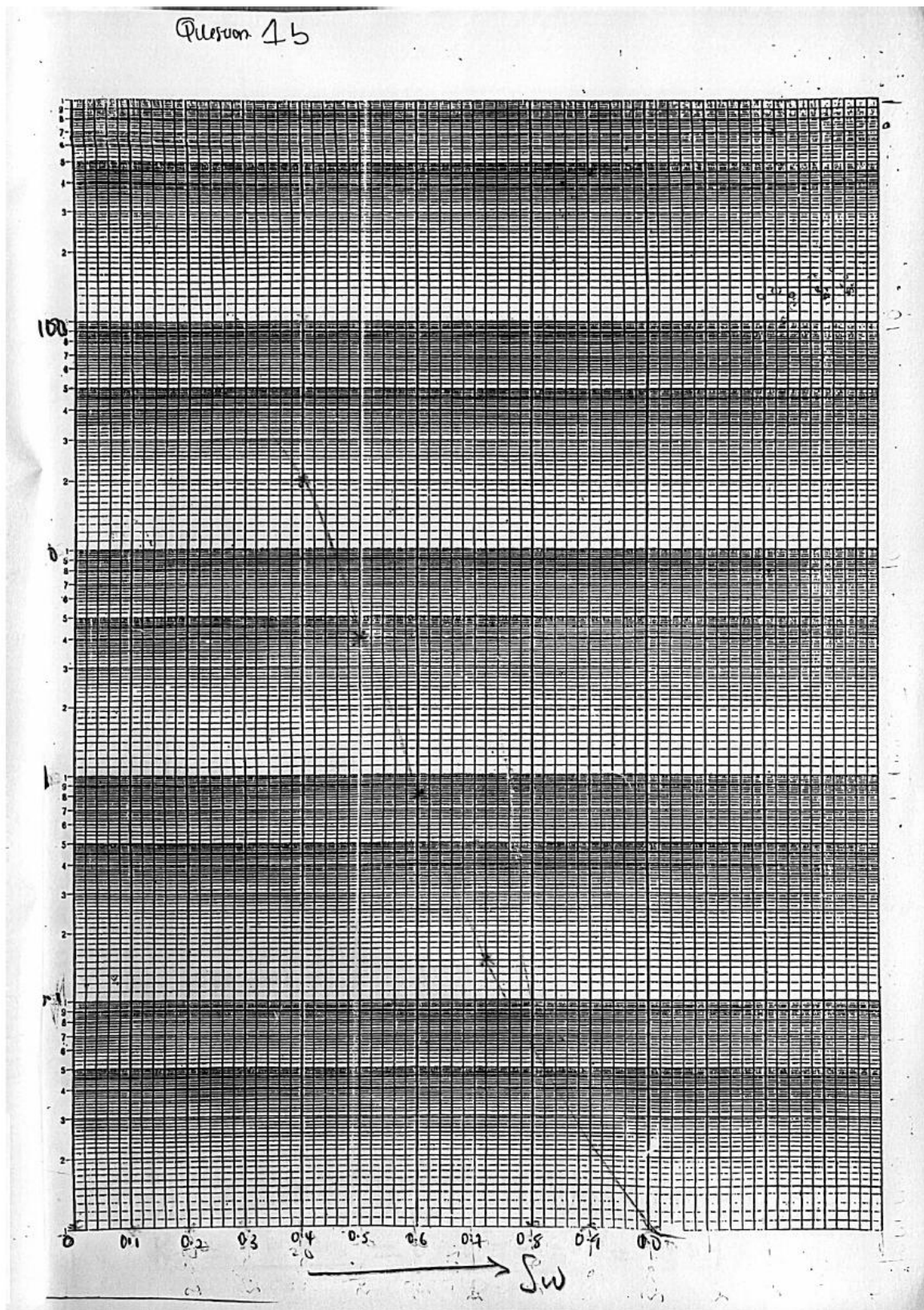
$$\beta_w = 1.05 \text{ bbl/stb}$$

$$\text{At } S_w = 0.5 (50\%)$$

Therefore the surface water at $S_w = 50\%$

$$f_w = \frac{1}{1 + \frac{k_o}{k_{rw}} \cdot \frac{\mu_w \beta_o}{\mu_o \beta_o}}$$

$$\text{But } \frac{k_o}{k_{rw}} = a e^{-b S_w}$$



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$$f_w = \frac{1}{1 + q e^{-\frac{5f_w}{H_0} \frac{H_w B_w}{B_0}}}$$

$$f_w = \frac{1}{1 + 12.497 \cdot 63 e^{-0.5 \times 16.094 \left(\frac{0.68 \times 1.05}{2.4 \times 1.5} \right)}}$$

$$f_w = 0.5576 \approx 0.56$$

1e for the subsurface water at (Refemour).

$$f_w = \frac{1}{1 + \frac{k_{rw}}{k_{rw}} \cdot \frac{B_w H_w}{B_0 H_0}}$$

But $\frac{B_w}{B_0}$ is negligible, $\frac{k_{rw}}{k_{rw}} = q e^{-5f_w}$

$$f_w = \frac{1}{1 + 12.497 \cdot 63 e^{-16.094 \times 0.5 \left(\frac{0.68}{2.4} \right)}}$$

$$f_w = 0.4687 \approx 0.47$$

1f) Initial connate water = 30%

water intrusion above transition = 50% (meaning 50% of the pore space is included by water)

$$S_{o1} = 100 - S_{wi} = 100 - 30 = 70\%$$

$$S_{o1} = \text{Initial oil saturation} = 70\% = 0.7 \quad S_{wi} = 0.3$$

$$S_{or} = \frac{S_{wi}}{1 - S_{wi}} \times \frac{1 - S_{wi}}{S_{o1}}$$

$$S_{or} = \left[\frac{0.3}{1 - 0.3} \times \frac{1 - 0.3}{0.7} \right] = 0.428 \times 0.5714 = 0.2445 \approx 0.25$$

$$R_f = \frac{S_{o1} - S_{or}}{S_{o1}} = \frac{0.7 - 0.25}{0.7} = 0.643$$

Percentage of oil recovered = $0.643 \times 100 = 64.3\%$

Therefore 64.3% of oil in the sandstone reservoir above the transition zone will be recovered under high pressure water drive.

19) Average Gas Saturation, $S_g = 15\%$

B_{oi} = initial oil fut = 1.50 bbl/stb

$B_o = 1.35$ bbl/stb

$$Rf = \left[1 - \frac{S_g}{1 - S_{wi}} \right] \times \frac{B_o}{B_{oi}}$$

$$\begin{aligned} Rf &= 1 - \frac{0.15}{1 - 0.7} \times \frac{1.35}{1.50} \\ &= (1 - 0.214) \times 0.9 \\ &= 0.707 \end{aligned}$$

Oil Percentage recovered = $0.707 \times 100 = 70.7\%$

70.7% of oil in the included portion will be recovered

In Sandstone Por volume = 100%

The least permeable pore volume makes up 20% (0.2)

The most " " " makes up 25% (0.25)

To estimate Absolute Permeability

$$K \propto (PV)$$

for the least permeable pore channel = $K_1 \propto PV_1^2 \Rightarrow K_1 = 0.2^2 = 0.04$

for the most " " " = $K_2 = 0.25^2 = 0.06$

To calculate the absolute permeability by each pore channel

fraction due to least permeable pore channel: $\frac{K_1}{K_1 + K_2} = \frac{0.04}{0.04 + 0.06} = 0.39$

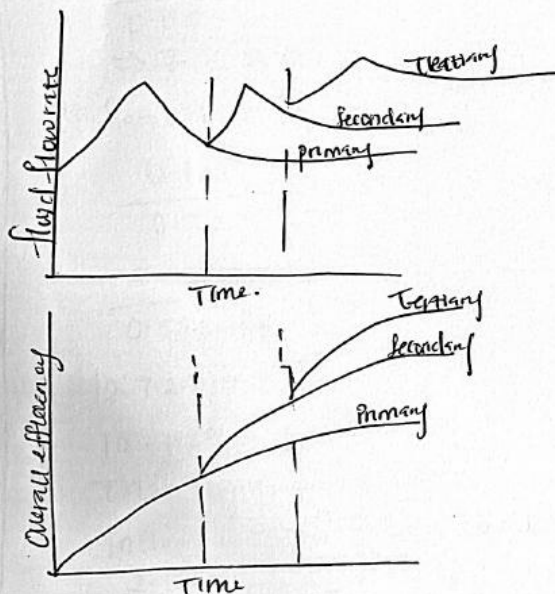
$$\therefore \text{fraction of the most permeable} = \frac{k_2}{k_1 + k_2} = \frac{0.0625}{0.04 + 0.0625}$$

$$\therefore 39\% \text{ of the absolute permeability is due to } = 0.61$$

least permeable pore channel while 61% of the absolute permeability is due to the most permeable pore channel.

Enhanced Oil Recovery also called Tertiary oil recovery is a technique used to recover oil and above what could be recovered by Primary and Secondary Method.

TARGET AND PATH OF EOR



- To maximize Recovery from oil and gas fields
- To maintain economic oil rate (increased profitability)
- To Reduce dependency on new exploration
- To meet the global energy demand.

2c) $\bar{S}_0$	$\bar{S}_w = 1 - \bar{S}_0$	$E_D = \frac{S_{w0} - S_{w1}}{1 - S_{w1}}$
0.65	0.35	0.1333
0.60	0.40	0.2000
0.55	0.45	0.2666
0.50	0.50	0.3333
0.35	0.65	0.5333 (maximum)

$$\bar{S}_0 = \frac{65}{100} = 0.65, \bar{S}_{0(2)} = \frac{60}{100} = 0.60, \bar{S}_0 = \frac{55}{100} = 0.55, \bar{S}_{0(4)} = \frac{50}{100} = 0.50$$

$$\bar{S}_{0(3)} = \frac{35}{100} = 0.35$$

Initial water saturation = 0.75

$$\bar{S}_{w1} = 1 - 0.75 = 0.25$$

$$\bar{S}_{w2} = 1 - 0.60 = 0.40$$

$$\bar{S}_{w(3)} = 1 - 0.55 = 0.45$$

$$\bar{S}_{w(4)} = 1 - 0.50 = 0.50$$

$$\bar{S}_{w(5)} = 1 - 0.35 = 0.65$$

$$E_D = \frac{\bar{S}_w - S_{w1}}{1 - S_{w1}}$$

$$\textcircled{1} E_D \text{ at } \bar{S}_0 = 0.65, S_{w0} = 0.35, S_{w1} = 0.25$$

$$\textcircled{1} E_D = \frac{0.35 - 0.25}{1 - 0.25} = 0.1333$$

$$\textcircled{2} \text{ At } S_{w0} = 0.40, E_D = \frac{0.40 - 0.25}{1 - 0.25} = 0.2000$$

$$\textcircled{3} \text{ At } S_{w0} = 0.45, E_D = \frac{0.45 - 0.25}{1 - 0.25} = 0.2666$$

$$\textcircled{4} \text{ At } S_{w0} = 0.50, E_D = \frac{0.50 - 0.25}{1 - 0.25} = 0.3333$$

③ At $S_w = 0.65$ $\Rightarrow \frac{0.65 - 0.25}{1 - 0.25} = 0.5333$

S_w	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.85	0.9	1
k_{ro}	1	0.99	0.98	0.62	0.35	0.2	0.12	0.06	0.05	0	0	0
k_{rw}	0	0	0	0.02	0.05	0.12	0.22	0.35	0.51	0.6	0.71	1

$$\frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$$

At $S_w = 0.4$, $k_{ro} = 0.35$, $k_{rw} = 0.05$

$$\frac{0.35}{0.05} = a e^{-b(0.4)}$$

$$\Rightarrow 7 = a e^{-b(0.4)} \rightarrow \text{①}$$

At $S_w = 0.6$, $k_{ro} = 0.12$, $k_{rw} = 0.22$

$$\frac{0.12}{0.22} = 0.55 \therefore 0.55 = a e^{-b(0.6)} \rightarrow \text{②}$$

$$7 = a e^{-b(0.4)}$$

$$0.55 = a e^{-b(0.6)}$$

$$12.727 = e^{-0.4b + 0.6b}$$

$$12.727 = e^{0.2b}$$

Take log of both sides

$$\ln(12.727) = \ln e^{0.2b}$$

$$2.543 = 0.2b$$

$$\therefore b = 12.718$$

Substitute in equation ①

$$7 = a e^{-b(0.4)}$$

$$7 = a e^{-(12.718)(0.4)}$$

$$a = \frac{7}{e^{5.0872}} = 0.648221133550$$

3b Given the following parameter
 $\mu_o = 0.2 \text{ cP}$ $\mu_w = 0.75 \text{ cP}$ $\beta_o = 1.65 \text{ rb/stb}$ $\beta_w = 1.05 \text{ rb/stb}$

Therefore the surface water cut with water saturation at 53% (0.53).

$$f_w = \frac{1}{1 + \left(\frac{k_{ro}}{k_{rw}} \right) \cdot \left(\frac{\mu_o \beta_o}{\mu_w \beta_w} \right)}$$

But $\frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$

$$\therefore f_w = \frac{1}{1 + a e^{-b S_w} \left(\frac{\mu_o \beta_o}{\mu_w \beta_w} \right)} = \frac{1}{1 + 1133.55 e^{-0.53 \times 12.718} \left(\frac{0.2}{0.75} \right)}$$

$$f_w = 0.7747 \approx 0.77$$

3c Reservoir cut (subsurface cut)

$$f_w = \frac{1}{1 + \left(\frac{k_{ro}}{k_{rw}} \right) \left(\frac{\beta_o \mu_o}{\beta_w \mu_w} \right)}$$

$\frac{\beta_o \mu_o}{\beta_w \mu_w}$ is negligible at reservoir condition.

$$f_w = \frac{1}{1 + 1133.55 e^{-0.53 \times 12.718} \left(\frac{1.05}{1.65} \right)}$$

$$f_w = 0.539 \approx 0.54$$

3d Factors responsible affecting recovery

- ① Capillary pressure
- ② Viscosity
- ③ Interfacial tension
- ④ Gravity
- ⑤ Capillary forces.

false, because any influence that causes the fractional flow curve to shift upward (ie increase in f_w or f_g) will result in a less efficient displacement process.

2) Oil recovery operations are divided into:

Primary recovery: It is a process of recovering (production) hydrocarbon from a reservoir by allowing the oil to flow naturally to the bottom of the well due to gravity and difference in pressure (pressure gradient) between the reservoir and the bottom of the well.

Secondary recovery: It involves producing hydrocarbon from the reservoir by increasing the natural energy of the reservoir, usually by injecting fluid into the reservoir to maintain or boost pressure (pore) and thereby pushing oil toward the producing well.

Tertiary recovery: This is a process of recovering of the remaining oil in the reservoir after a conventional primary and secondary recovery project (phases) such as water-flooding. It is also known as Enhanced oil recovery.

3) Examples.

Primary recovery — Solution gas drive, water drive, Rock and liquid expansion.

Secondary recovery — Water flooding, Gas flooding

Tertiary recovery — Surfactant flooding (Chemical flooding) / Thermal fluids

ic) 4-fluids used in EOR.

1) Water

Objective: It maintains reservoir pressure and displace oil towards production well through waterflooding.

2) CO₂

Objective: Reduces oil viscosity and improve oil displacement by making the oil less thick, enhancing recovery in reservoir.

Surfactant: It reduces the interfacial tension between oil and water, improves oil mobility

4a EOR involves injecting fluids into an oil reservoir that increases oil recovery over that which would be achieved using natural energy of reservoir [natural drive mechanism].

Examples: Polymer flooding, foam injection, miscible gas injection or water alternating gas [WAG].

While

IDR is a term that applies to improvement in oil recovery achieved better engineering and project management.

Examples: Four dimensional seismic surveying, Improved completion, New capacity to drill deviated, horizontal and multilateral wells or Development in measurement while drilling.

4b) FOUR PRACTICAL CONSIDERATIONS FOR EOR DESIGNS

- ① Selection of flooding pattern
- ② Economic consideration (cost)
- ③ Surface facilities and infrastructure.
- ④ Reservoir characterization and modelling.

4c) The displacement of oil by water in stratified reservoirs is a key in secondary oil recovery referred to as waterflooding. In stratified the rock strata have different permeability and porosity characteristics which affects how water can displace oil.

4d) The ultimate recovery factor is different for different oil reservoirs because of their different reservoir characteristics and different recovery mechanism.

4e) Fractional flow of a fluid represents the proportion of the total fluid flow rate that is comprised of a specific fluid phase / as oil / water / or gas.

Polymer Solutions.

Objective: It increase water viscosity to improve sweep efficiency.

Gas lifting is a principle under primary recovery method. It is an Artificial lift process involving injecting high pressure gas to supplement reservoir pressure and lift the crude oil to the surface (It reduces crude viscosity).

2022/2023
PEE/PNG 511 - ENHANCED OIL RECOVERY

QUESTION 1

A) Enhanced oil recovery sometimes referred to as tertiary oil or improved oil recovery is defined to include all processes that increase oil recovery beyond primary and secondary recovery processes.

This definition was broadened to include all EOR processes that have been used to improve oil recovery beyond the primary stage of reservoir depletion or at the end of secondary recovery processes involving water flooding or it is applied to a field that is experiencing high water cut and depletion level but still have large pockets of oil left unswept (residual oil). The target of many EOR technologies is to extract residual oil which amounts to about 2-4 trillion barrels (Hall et al, 2003) or about 67% of the total world oil reserves (Bryant et al, 1993).

B

Companies embark on EOR activities for numerous reasons including to;

- Maximize the amount of oil reserves recovered
- Extend the life of oil fields.
- Increase recovery factor of a field
- Helps increase domestic oil supplies while also providing a way to safely and permanently store CO_2 underground.

IC

The terms improved oil recovery (IOR) and enhanced oil recovery (EOR) are oftentimes overlapping in meaning and are sometimes used interchangeably.

However, improved oil recovery involves methods for recovery additional oil beyond primary recovery technique (fluid expansion, rock compressibility, gravity drainage, pressure decline and natural waterdrive or gasdrive). The term improved oil recovery is used both in a restricted sense and a more generic sense. In its restricted sense, it is a process, such as water flooding or gas flooding, that adds energy to a reservoir to increase recovery factor. In its more general sense, it is any activity that increases oil production and recovery factor which may include EOR, infill drilling, hydraulic fracturing and drilling horizontal and multilateral wells.

In a nutshell, any additional method to increase oil production beyond primary recovery (natural reservoir energy) is termed improved oil recovery.

On the other hand, EOR is a recovery method that uses sophisticated techniques that alter the original properties of oil. Once ranked as a third stage of oil recovery that was carried out after primary and secondary recovery. Its purpose is not only to restore formation pressure, but also to improve oil displacement or fluid flow in the reservoir.

Examples of IOR processes;

- Water flooding
- Gas flooding
- Water/gas injection
- Infill drilling
- Hydraulic fracturing
- Drilling horizontal and multilateral wells.

Examples of EOR processes;

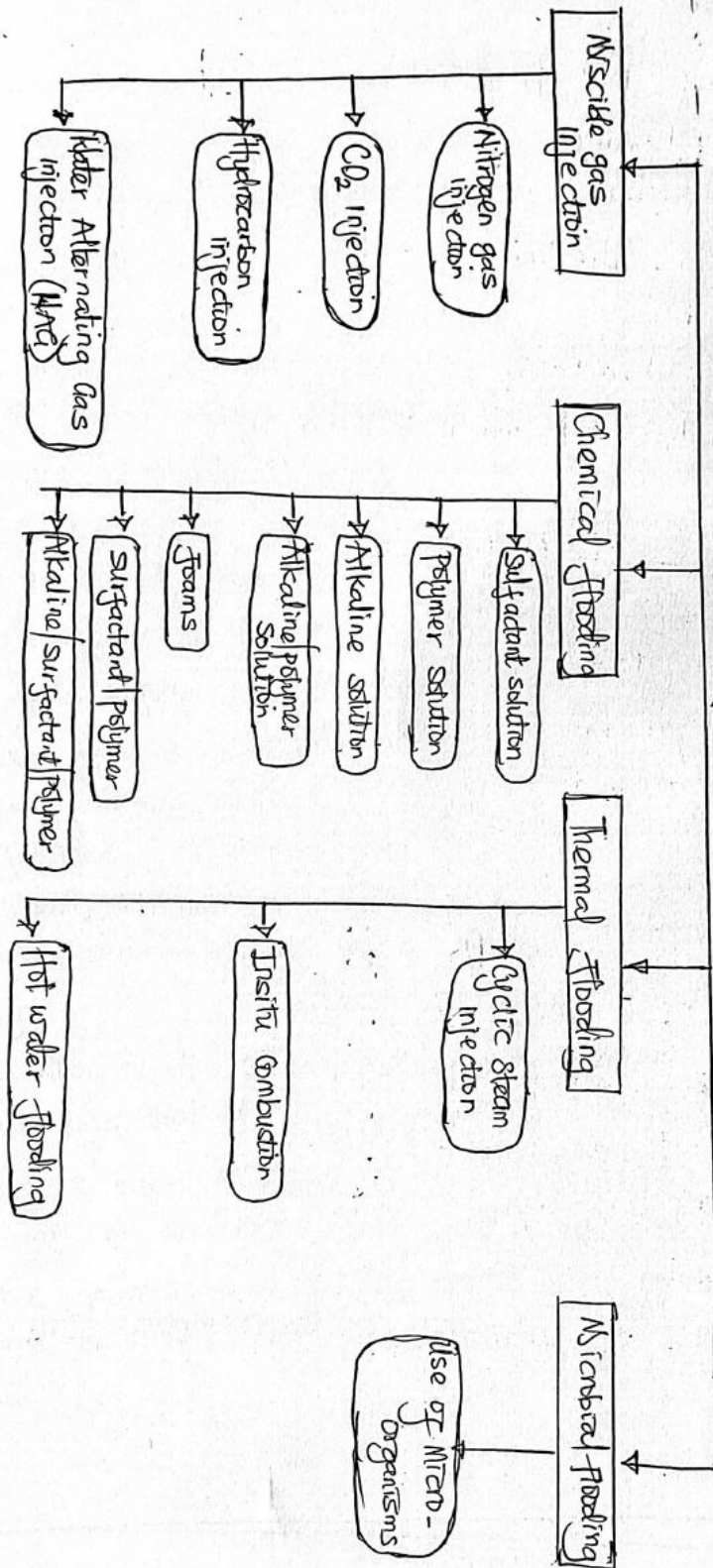
- Miscible gas injection
- Chemical flooding
- Thermal flooding
- Microbial flooding

1D

Considerations for EOR include;

- Reservoir geometry and heterogeneity
- Fluid saturation and properties
- Primary recovery drive mechanism
- Reservoir depth.
- Availability of resources for the preferred EOR type.

1. EOR (Enhanced Oil Recovery) Processes



QUESTION 2

A) The fractional flow of a fluid 'x' in the presence of another immiscible fluid 'y' is defined as the flowrate of the fluid x divided by the total flowrate in the system.

For the fractional flow of water in the presence of an immiscible fluid such as oil, the equation is given by;

$$f_w = \frac{q_w}{q_t} = \frac{q_w}{q_w + q_o}$$

The development of this equation is attributed to Leverett in 1941. The fractional flow of water is also termed "water cut".

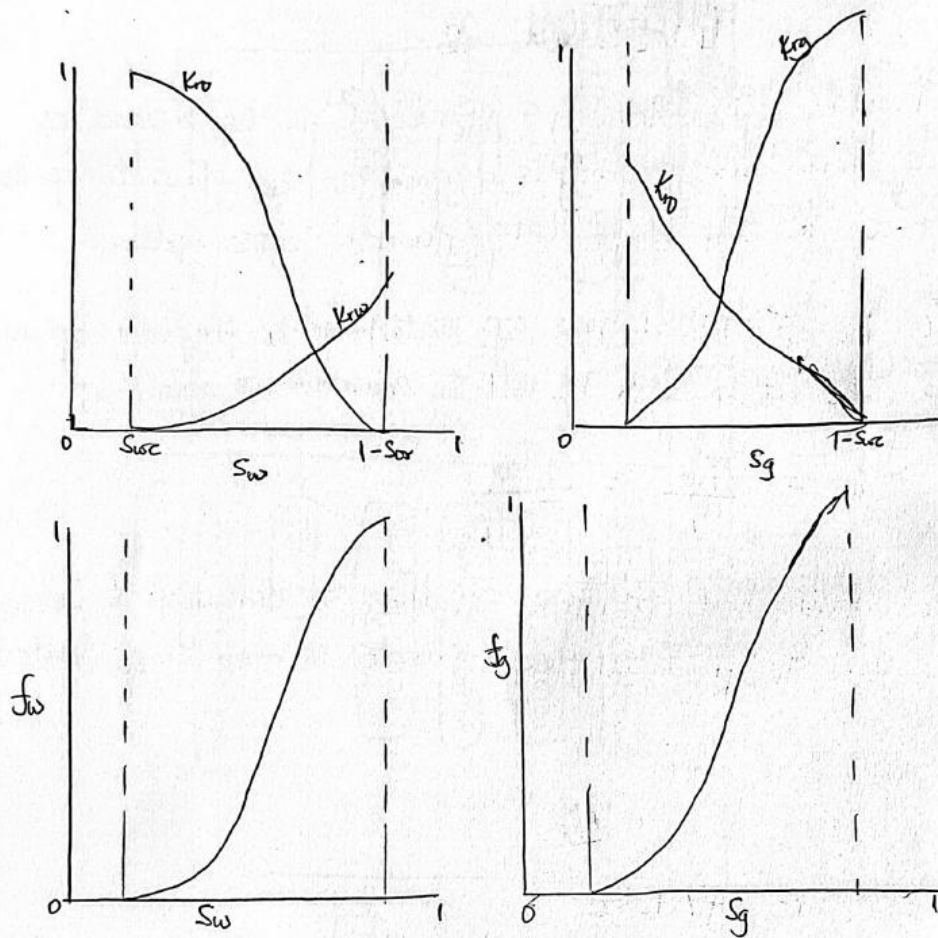
2B

Answer is FALSE.

This is because from the definition of water cut i.e. $f_w = \frac{q_w}{q_w + q_o}$, we notice that the limits of the water cut

are 0% and 100%. At the irreducible water saturation, the water flowrate q_w is zero and, therefore, the water cut is 0%. At the residual oil saturation point, S_{or} , the oil flowrate is zero and the water cut reaches its upper limit of 100%.

The shape of f_w Vs S_w curve is characteristically S-shaped as shown below;



From the diagram, it can be observed that the limits of the f_w curve (0 and 1) are defined by the end points of the relative permeability curves.

Therefore, in general, any influences that cause the fraction flow curve to shift upward (i.e. increase in f_w or f_g) will result in a less efficient displacement process. ~~It is less~~ Because during the displacement of oil by water flood, an increase in f_w at any point in the reservoir will cause a proportional decrease in f_o and oil mobility.

* At oil saturation of 60%; $\bar{S}_o = 60$ or 0.6
 $\bar{S}_w = 1 - 0.6 = 0.40$

$$F_D = \frac{0.4 - 0.25}{1 - 0.25} = 0.20$$

* At oil saturation of 55%; $\bar{S}_o = 0.55$
 $\bar{S}_w = 1 - 0.55 = 0.45$

$$F_D = \frac{0.45 - 0.25}{1 - 0.25} = 0.2667$$

* At oil saturation of 50%; $\bar{S}_o = 0.5$
 $\bar{S}_w = 1 - 0.5 = 0.5$

$$F_D = \frac{0.5 - 0.25}{1 - 0.25} = 0.3333$$

* At oil saturation of 35% (residual), $S_{or} = 0.35$
 $\bar{S}_w = 1 - 0.35 = 0.65$

$$F_D = \frac{0.65 - 0.25}{1 - 0.25} = 0.5333$$

Therefore; the results is represented on the table below;

$\bar{S}_o$	$\bar{S}_w = 1 - \bar{S}_o$	$F_D = \frac{\bar{S}_w - S_{wi}}{1 - S_{wi}}$
0.65	0.35	0.1333
0.60	0.40	0.20
0.55	0.45	0.2667
0.50	0.50	0.3333
$S_{or} = 0.35$	0.65	0.5333 (maximum)

QUESTION 3

Given the table and the parameters in the question;
 Firstly, calculate ^{the reservoir} fractional flow for each saturation using the equation;

$$f_w = \frac{1}{1 + \frac{K_{ro}}{K_{rw}} \frac{\mu_w}{\mu_o}} \quad \text{where } \mu_w \text{ \& } \mu_o \text{ are 1 and 2 cp respectively}$$

Neglecting capillary term and given $\alpha = 0$

Therefore the table becomes;

S_w	K_{ro}/K_{rw}	$f_w = \frac{1}{1 + \frac{K_{ro}}{K_{rw}} \frac{\mu_w}{\mu_o}}$
0.25	30.23	0.0621
0.30	17.00	0.1053
0.35	9.56	0.1730
0.40	5.38	0.2710
0.45	3.02	0.3984
0.50	1.70	0.5405
0.55	0.96	0.6757
0.60	0.54	0.7874
0.65	0.30	0.8696
0.70	0.17	0.9217
0.75	0.10	0.9524

Next, plot a graph of f_w Vs S_w on a Cartesian Scale. (Graph is attached behind)

From the graph;

$$(S_{wf}, f_{wf}) = (0.596, 0.78) \text{ and } \left(\frac{df_w}{dS_w} \right)_{surf} = 1.972$$

From the question;

When the saturation reaches 0.70 (i.e. $S_{w2} = 0.7$)

At water saturation of 0.7, the corresponding water cut (f_{w2}) = 0.9217 $\approx$ 0.922

$\therefore$

A) Reservoir cut = 0.922

B) Surface water cut can be estimated using the eqn

$$f_{ws} = \frac{B_o}{B_w \left(\frac{1}{f_w} - 1 \right) + B_o}$$

$$\begin{aligned} f_{ws} &= \frac{1.25}{1.02 \left(\frac{1}{0.922} - 1 \right) + 1.25} \\ &= 0.9854 \text{ STB/STB} \end{aligned}$$

C) Producing reservoir WOR is given by;

$$WOR_r = \frac{f_w}{1 - f_w} = \frac{0.922}{1 - 0.922}$$

$$WOR_r = 11.8205 \text{ bbl/bbl}$$

D) Surface-water-oil ratio is given by;

$$kOR_s = \frac{B_o kOR_r}{B_w}$$

$$kOR_s = \frac{1.25 \times 11.8205}{1.02} = 14.4859 \text{ STB/STB}$$

E) Average water saturation in the swept Area. ($\bar{S}_{w2}$)

$$\bar{S}_{w2} = S_{w2} + \frac{1 - f_{w2}}{\left(\frac{df_w}{dS_w}\right)_{S_{w2}}}$$

To get $\left(\frac{df_w}{dS_w}\right)_{S_{w2}}$, draw a tangent to the fractional flow curve at the co-ordinate $(S_w, f_w) = (0.7, 0.922)$ and extrapolate to $f_w = 1$. Equivalently, the average water saturation can be calculated by determining the slope of the tangent

$$\text{From the graph; } \left(\frac{df_w}{dS_w}\right)_{S_{w2}} = \frac{f_{w2} - f_{w1}}{S_{w2} - S_{w1}} = \frac{1 - 0.834}{0.8 - 0.6} = 0.83$$

$$\therefore \left(\frac{df_w}{dS_w}\right)_{S_{w2}} = 0.83$$

and;

$$\bar{S}_{w2} = 0.7 + \frac{1 - 0.922}{0.83} = 0.7940$$

f) pore volumes of water injected.

$$Q_i = \frac{1}{\left(\frac{df_w}{ds_w}\right)_{sw2}}$$

$$\therefore Q_i = \frac{1}{0.83} = 1.2048$$

g) Cumulative water injected:

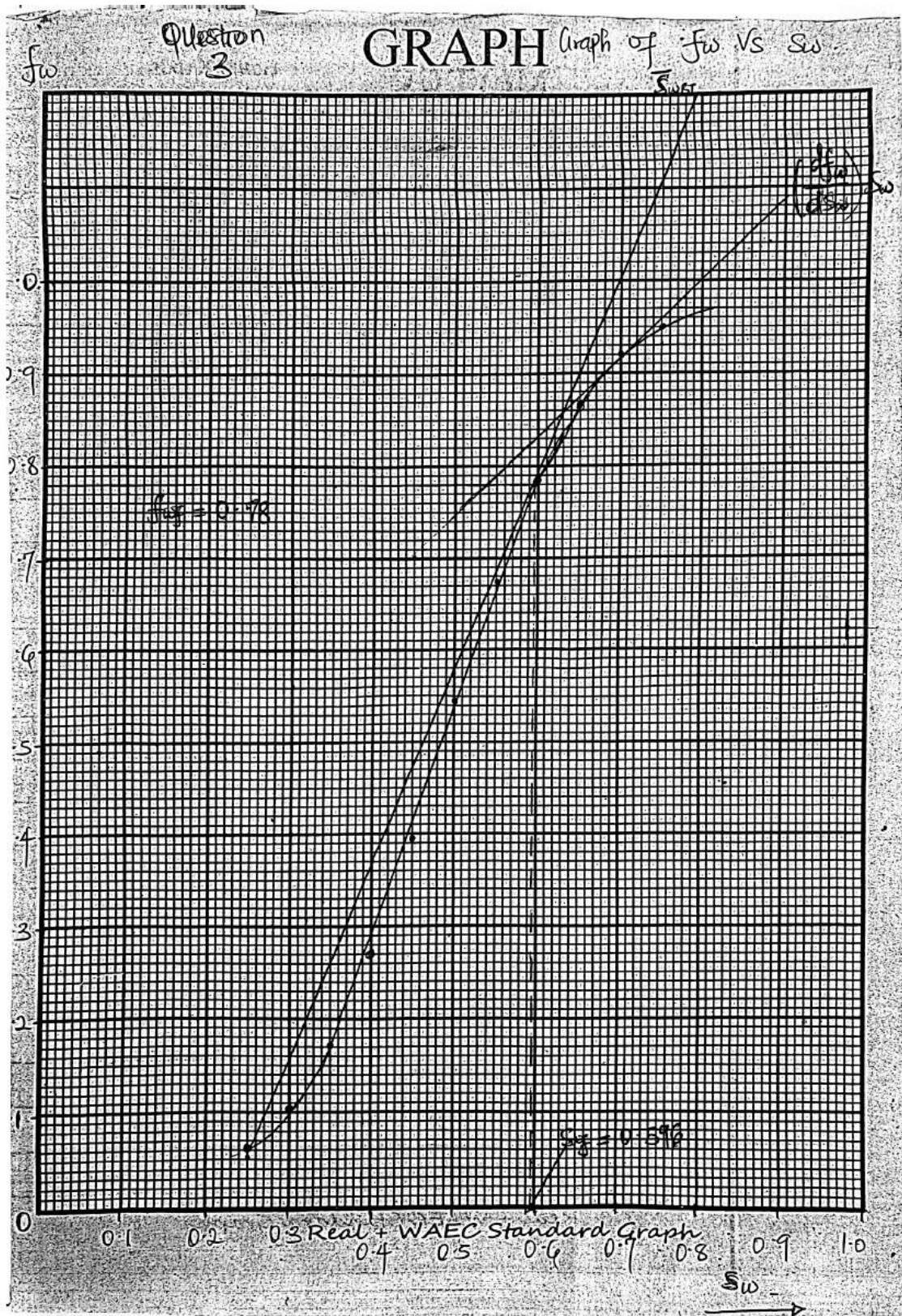
$$W_{inj} = PV(Q_i)E_A E_v$$

where $E_A = E_v = 1$

$$\text{and } PV = \frac{\phi A L}{5.615} = \frac{0.25 \times 26400 \times 600}{5.615}$$

$$PV = 705253.7845 \text{ bbls}$$

$$\therefore W_{inj} = 705253.7845 \times 1.2048 \times (1)(1) \\ = 849689.7596 \text{ bbl}$$



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QUESTION FOUR (4)

A) For the solution to 4a,

REFER TO 2019/2020 solution (Question 1A)

B) The principal objective of polymer augmented waterflooding on an oil reservoir is to increase the oil-production rate and, ultimately, the oil recovery. This is accomplished by "voltage replacement" - Injection of water to increase the reservoir pressure to its initial level and maintain it near that pressure.

Other objectives of this recovery mechanism include;

- To control the mobility of the wetting phase.
- To improve macroscopic sweep in a waterflood to viscousify the water.
- To recover substantial increment of the oil in place, typically 80% more.
- To achieve favourable fluid mobility ratio

4c

Mobility ratio can be defined as the mobility of an injectant (injected fluid) divided by that of the fluid it is displacing, such as oil. In summary, it is the ratio of the displacing fluid to the displaced fluid.

48

In a displacement process, a mobility ratio is considered unfavourable if it is greater than 1.

In this case, the injected fluid may bypass a section of the reservoir known as a phenomenon termed Viscous Fingering.

QUESTION FIVE

Given the permeability vs saturation data below; The relative permeability ratio is derived thus;

S_w	K_{ro}	K_{rw}	K_{ro}/K_{rw}
0	1.0	0	∞
10	1.0	0	∞
20	1.0	0	∞
30	0.94	0	∞
40	0.8	0.04	20
50	0.44	0.11	4
60	0.16	0.2	0.8
70	0.045	0.3	0.15
75	0	0.36	0
80	0	0.44	0
90	0	0.68	0
100	0	1.0	0

5A) The graph of the relative permeabilities of oil and water against water saturation is shown in the Cartesian graph attached.

5B) The graph of the relative permeability ratio of oil and water Versus water saturation is shown in the Semi-Log Plot graph attached.

5c) From the graph;

b = slope and a = intercept.

$\therefore$ calculating slope from graph;

$$b = \frac{\left(\frac{k_{ro}}{k_{rw}}\right)_2 - \left(\frac{k_{ro}}{k_{rw}}\right)_1}{S_{w2} - S_{w1}} = \frac{\ln(20) - \ln(4)}{0.5 - 0.4} = 16.0944$$

The intercept can be derived from the equation;

~~$$\frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$$~~

$$\frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$$

At $\frac{k_{ro}}{k_{rw}} = 4$, $S_w = 0.5$

$$\therefore 4 = a e^{-16.0944(0.5)}$$

$$a = \frac{4}{e^{-16.0944 \times 0.5}} = 12500.1305$$

Using two pivot points and solving simultaneously;

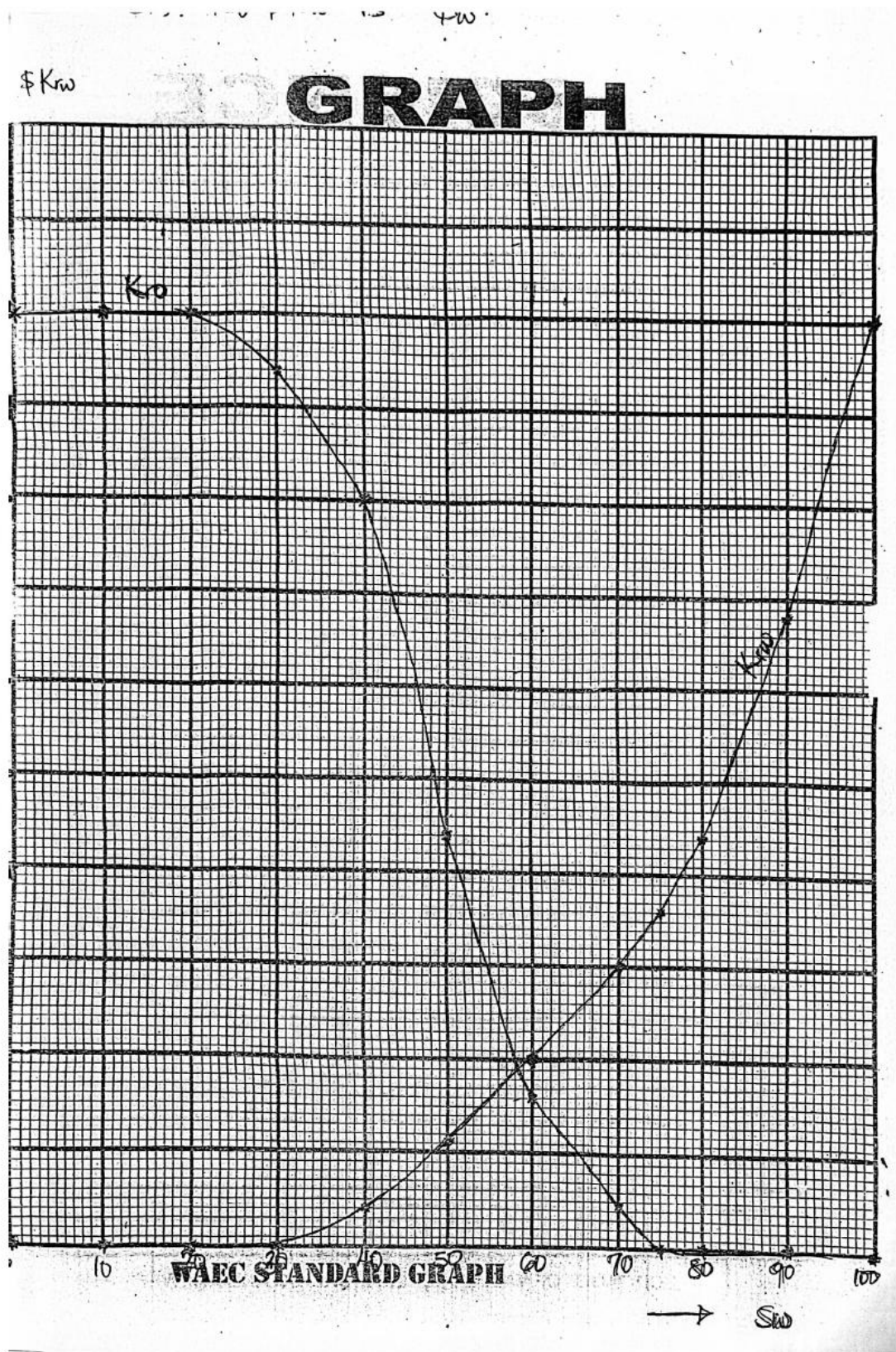
Using pivot points of 50 and 60%

At saturation of 50% i.e. $S_w = 0.5$, $\frac{k_{ro}}{k_{rw}} = 4$

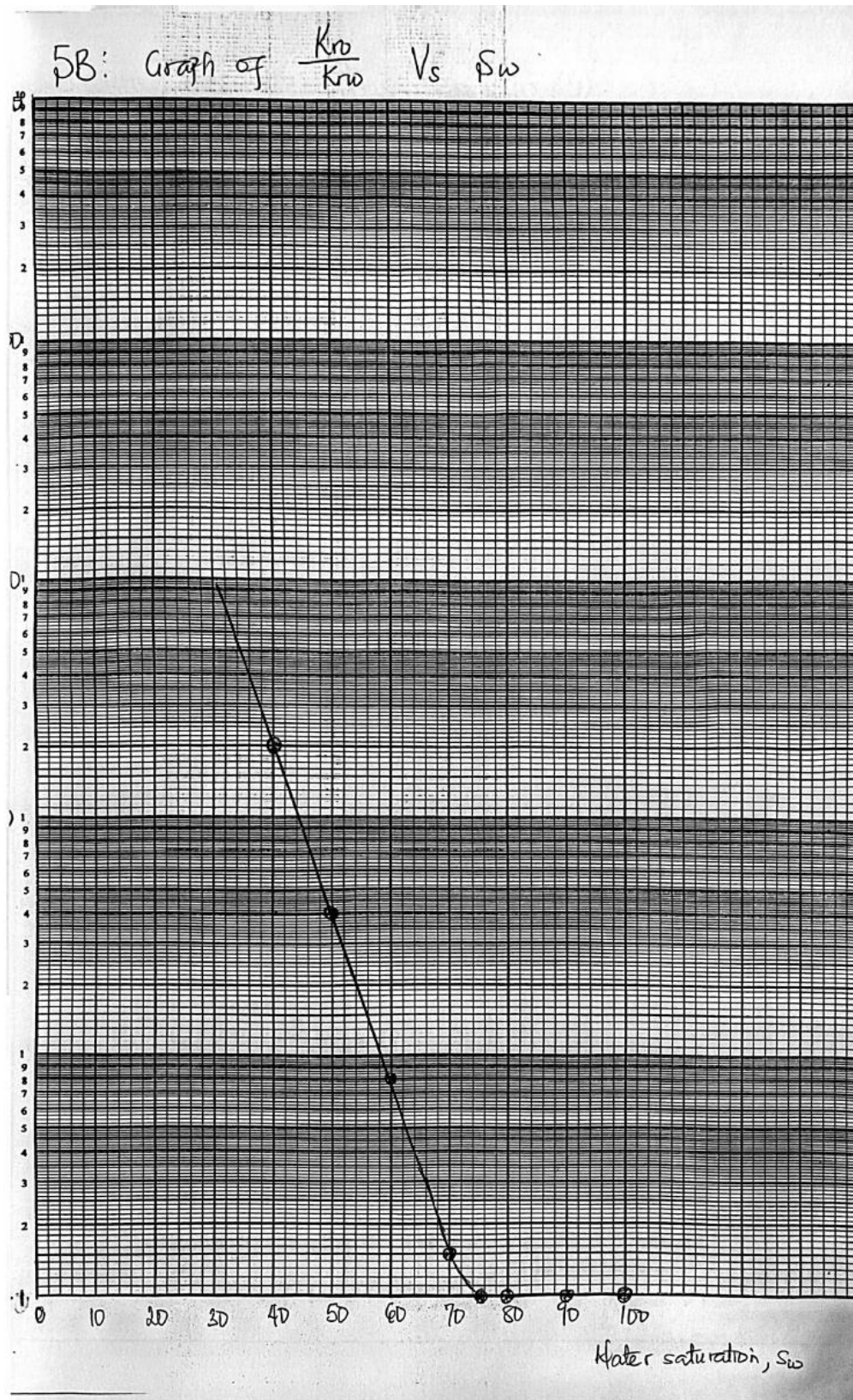
$$\therefore \frac{k_{ro}}{k_{rw}} = a e^{-b S_w} \quad \text{; } 4 = a e^{-0.5b} \quad \dots \dots \dots (i)$$

At saturation of 60% i.e. $S_w = 0.6$, $\frac{k_{ro}}{k_{rw}} = 0.8$

$$\therefore 0.8 = a e^{-0.6b} \quad \dots \dots \dots (ii)$$



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Dividing equation (i) by equation (ii)

$$\frac{4}{0.8} = \frac{ae^{-0.5b}}{ae^{-0.6b}}$$

$$5 = e^{-0.5b - (-0.6b)}$$

$$5 = e^{0.1b}$$

$$\ln(5) = 0.1b$$

$$b = \frac{1.6094}{0.1} = 16.094$$

Therefore substituting $b = 16.094$ in eqn (i)

$$4 = ae^{-0.5b}$$

$$4 = ae^{-0.5(16.094)}$$

$$a = \frac{4}{e^{-0.5(16.094)}} = 12497.6307$$

(5D) Given $\mu_o = 3.4 \text{ cp}$, $\mu_w = 0.68 \text{ cp}$, $\beta_o = 1.506 \text{ bbl/STB}$, $\beta_w = 1.056 \text{ bbl/STB}$, water saturation at transition zone, $S_w = 55\%$

Surface water cut (f_{ws}):

$$f_{ws} = \frac{1}{1 + \frac{K_{ro} \mu_w \beta_w}{K_{rw} \mu_o \beta_o}} = \frac{1}{1 + ae^{-bS_w} \frac{\mu_w \beta_w}{\mu_o \beta_o}}$$

$$f_w = \frac{1}{1 + 12497.6307 e^{-16.094 \times 0.55} \left[\frac{0.68 \times 1.05}{3.4 \times 1.5} \right]}$$

$$f_w = \frac{1}{1.56} = 0.6410 \text{ or } 64.10\%$$

5E) Reservoir out;

$$f_w = \frac{1}{1 + qe^{-bS_w} \frac{\mu_o}{\mu_w}} = \frac{1}{1 + (12497.6357) e^{-16.694 \times 0.5} \times \frac{0.68}{3.4}}$$

$$f_w = 0.5556 \text{ or } 55.56\%$$

5F) Percentage recovery above transition zone;

Water saturation when there is no longer oil movement;

At invaded zone of water; $\bar{S}_o = 0.25$ and $\bar{S}_w = 0.75$

$\therefore$ Given $S_{wi} = 30\%$ i.e. 0.3

$$\% \text{ Recovery} = \frac{\bar{S}_w - S_{wi}}{1 - S_{wi}} = \frac{0.75 - 0.3}{1 - 0.3} = 0.642857 \approx 64.2857\%$$

5A) At pressures below saturation pressure;

$$S_{gi} = 15\%$$

$\therefore$ At saturation pressure, $S_{oi} = 1 - S_{wi} - S_{gi}$

$$S_{oi} = 1 - 0.3 - 0.15 = 0.55$$

However, in the swept area, the gas saturation is considered zero, thus

$$\bar{S}_o = 1 - \bar{S}_w = 1 - 0.75 = 0.25$$

Given B_o at invaded area = 1.35 bbl/STB

$$\% \text{ Recovery} = \frac{\frac{S_{oi}}{B_{oi}} - \frac{S_o}{B_o}}{\frac{S_{oi}}{B_{oi}}}$$

$$= \frac{\frac{0.55}{1.5} - \frac{0.25}{1.35}}{\frac{0.55}{1.5}}$$

$$= 0.4949 = 49.49\% \\ \approx 50\%$$

5H)

QUESTION SIX (6)

6A.

The displacement of oil by water in a stratified reservoir refers to the process by which water is injected (water flood) into the reservoir and displaces the oil as a result of a density difference leading to a change in the fluid distribution within the reservoir. Stratified reservoirs are typically multiple layers of reservoir rocks each having different fluid types and distinct rock ^{& fluid} properties including density, viscosity and permeability.

For a water-wet rock, displacement of oil by water increases the saturation of the wetting phase and the fluid mobility increases with an increase in saturation. When water is injected into a stratified reservoir through injection wells, it displaces oil into producer wells through various displacement mechanisms including viscous displacement, gravity drainage and capillary imbibition.

Water after being treated to remove any material that might interfere with the movement in the reservoir, is injected through some wells in the field. It then moves through the formation, pushing oil towards the remaining production wells. The injection wells are usually located in a pattern that will best push oil towards the production wells.

Water injected into a stratified system will preferentially enter the layers of highest permeability moving with higher velocity. Consequently, the time of breakthrough in high-perm zones, a significant fraction of the less-permeable zones will remain unflooded.

QB

Sweep efficiency is a concept used to measure the effectiveness of water flooding in displacing oil within the reservoir.

Sweep efficiency measures how effectively the displacing fluid displaces oil within the reservoir. It quantifies the percentage of the oil reservoir volume that has been swept or contacted by the injected water leading to oil displacement.

Generally ~~the~~ sweep efficiency is divided into displacement efficiency, Areal sweep and Vertical sweep efficiency. The product of these efficiencies give the recovery factor

$$RF = E_D E_A E_V$$

The displacement efficiency E_D is the fraction of movable oil that has been displaced from the swept zone at any given time or pore volume injected. E_D is always less than 1

The areal sweep efficiency E_A is the fractional area of the pattern that is swept by the displacing fluid. The factors determining areal sweep are fluid mobilities, pattern type, Areal heterogeneity, Total volume of fluid injected.

The vertical sweep efficiency E_V is the fraction of the vertical section of the pay zone that is contacted by injected fluid. E_V is primarily a function of vertical heterogeneity, degree of gravity segregation, fluid mobilities and total volume injected.

Note that the product of $E_A E_V$ is called the Volumetric Sweep efficiency.

6C

Factors affecting water cut:

- Injection and production rate
- Water and oil viscosities
- Dip angle
- Reservoir Area and geometry
- Capillary pressure
- Reservoir heterogeneity
- Fluid mobility.

6D

Immiscible fluid displacement refers to a process where a fluid that is immiscible or only partially miscible with the reservoir fluid is injected into the reservoir to displace and recover residual oil. This technique is employed when the injected fluid typically water or surfactant does not mix with the reservoir oil.

Therefore, the injected fluid displaces the oil and pushes it towards the producer wells. This displacement relies on ~~grav~~ density and viscosity differences between the fluids, interfacial tension and capillary pressure to enhance displacement of oil. However, a number of factors should be considered including the optimum time of injection in order to prevent the injected fluid bypassing a section of the reservoir and thus preventing displacement (fingering).

Water flooding an oil reservoir is an example of an immiscible fluid displacement process.

QUESTION 7

Given the following parameters;

$$L = 10\text{cm}, A = 2\text{cm}^2, q = 0.008\text{cm}^3/\text{sec}, \mu = 2.5\text{cp},$$

$$\Delta p = 1.5\text{atm}$$

A) From Darcy Equation; Absolute permeability of the rock;

$$q = \frac{KA\Delta p}{\mu L}$$

$$K = \frac{q\mu L}{A\Delta p} = \frac{0.008 \times 2.5 \times 10}{2 \times 1.5}$$

$$= 0.0667\text{D} \text{ or } 66.7\text{mD}$$

B) Flowrate of brine of 0.75cp

$$q = \frac{KA\Delta p}{\mu} = \frac{0.0667 \times 2 \times 1.5}{0.75 \times 10}$$

$$q = 0.02668\text{cm}^3/\text{sec}$$

C) At 100% saturation, permeability equals Absolute permeability. This implies that the rock permeability would remain the same since rock permeability (absolute permeability) is independent of the type of fluid present as long as it occupies 100% of the conductive (effective ϕ) pore space.

7D.

Reservoir heterogeneity refers to the uneven changes in spatial distribution and internal properties of oil and gas reservoirs due to the influence of sedimentary environment, diagenesis and tectonic activities during the formation process.

In reservoir studies, reservoir heterogeneity probably has more influence than any other factor on the performance of a secondary or tertiary injection project. They affect sweep efficiencies, reservoir characterization, modelling and simulation.

The two most common types of heterogeneity are;

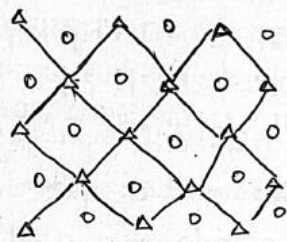
- Vertical heterogeneity
- Areal heterogeneity

Vertical heterogeneity is by far the most significant parameter influencing the vertical sweep and ~~the~~ variation in the vertical direction. A reservoir may exhibit many different layers in the vertical section that have highly contrasting properties.

Areal heterogeneity includes areal variation in permeability properties (h , K , ϕ , S_{wc}), geometrical factors, faults and boundary conditions due to the presence of an aquifer or gas cap.

7E

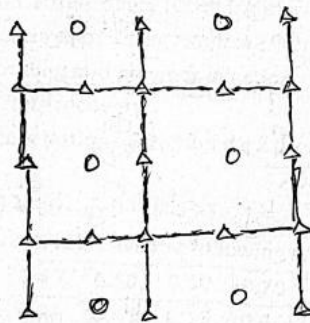
i) 5-spot normal: This is a special case of the staggered line drive in which the distance between all like well is constant i.e. $a = 2d$. Any four injection well thus forms square with a production well at the center.



5-spot Normal

Δ - injector wells
O - producer wells

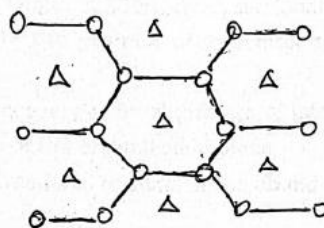
ii) 9-spot Normal: This pattern is similar to that of the five spot but with an extra injection well drilled at the middle of each side of the square. This pattern essentially contains eight injectors surrounding one producer.



9-spot Normal

Δ - injection wells
O - production wells

iii) Inverted seven spot: This is the opposite of the normal seven spot. The production wells are located at the corner of a hexagon with an injection well at its center.



Δ - injection wells
O - producing wells

7f.

i) **Polymer Flooding**: One means of achieving more favorable mobility ratio, and thus macroscopic sweep, in a water flood is to viscify the water. This is achieved by using water-soluble polymers of 2-propenamide and acrylic acid as the partly neutralized sodium salt in a ratio of about 70:30 polymer to acid by weight. Polymers are dissolved in the injection water, this raises its viscosity, thus reducing the mobility of the water. Polymer flooding will not only accelerate the recovery of oil but can also minimize viscous fingering in water flood projects.

ii) **Hydrocarbon miscible flooding**: This involves the use of hydrocarbon gases (Natural gas) that introduces miscible hydrocarbon gases into the reservoir. A miscible displacement process mixes with the oil to lower its viscosity, increase mobility, maintain reservoir pressure by reducing interfacial tension between oil and water.

For this method, LPG is used. It is more expensive than CO_2 & Nitrogen gas injection but doesn't strongly rely on reservoir temperature, pressure and crude oil composition.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT: PETROLEUM ENGINEERING

COURSE TITLE: EOR, CODE: PNG 511 TIME: 2hrs 30mins

Attempt any 4 question

1. The following permeability data were measured on a sand stone as a functions of its water saturation

S_w	0	10	20	30	40	50	60	70	75	80	90	100
K_{ro}	1.0	1.0	1.0	0.94	0.80	0.44	0.16	0.08	0	0	0	0
K_{rw}	0	0	0	0	0.04	0.11	0.20	0.30	0.36	0.44	0.68	1.0

*Critical saturation for oil and water.

- Find the constants "a" and "b" in the water saturation Vs related relative permeability equation $k_{ro}/k_{rw} = a e^{-b S_w}$ by substituting two set of data in the said equation and solved simultaneous (use water saturation values of 50% and 60% as pivot point).
- If $\mu_o = 2.4\text{cp}$, $\mu_w = 0.68\text{cp}$, $B_o = 1.5\text{rb/stb}$, and $B_w = 1.05\text{rb/stb}$, what is the surface water cut of a well completed in the transition zone where the water saturation is 50%.
- What is the reservoir cut in (b) above
- Explain why the ultimate recovery factor is different for different oil reservoirs
- Define producing reserves by energy source criteria
- What is polymer flooding and polymer injection
- List the forces affecting recovery
- From the Darcy flow equation, $(P_c - P_w) = q \times \mu \times B \times \ln(r_e/r_w)$, derive the formula for

$$7.08 k \times h$$

water cut (f_w) in a flow stream.

3. A linear sand stone reservoir have the following relative permeability characteristics

S_w	0.24	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
K_{ro}	0.15	0.88	0.74	0.45	0.19	0.06	0.0	0.0	0.0
K_{rw}	0.0	0.01	0.04	0.09	0.17	0.28	0.44	0.67	1.00

Other reservoir and fluid properties are as follows: Oil viscosity, $\mu_o = 1.25\text{cp}$, water viscosity $\mu_w = 0.75\text{cp}$, porosity $\phi = 0.25$, FVF $B_o = 1.12\text{bbl/stb}$, FVF $B_w = 1.03\text{bbl/stb}$

- If the average reservoir pressure is above the saturation pressures, develop a depth of the fractional flow of water as a function of water saturation. Assume the angle of deep is 0 and neglect capillary pressure.
- Prepare a graph of producing water oil ratio in surface unit as a function of average water saturation in the reservoir
- If the system has a length of 500ft, cross sectional area of 1ft^2 and water injection rate of $0.1\text{ft}^3/\text{hr}$. Determine the position of the displacement front from the inlet end after 100hrs.
- Determine the oil recovery at breakthrough of injected water at the producing end of the system as a fraction of the original oil in place
- What could be the fractional recovery if the abandonment water oil ratio is 25, 000STB.

- 4a. The data for an Okaka oil reservoir which is proposed for water flooding is present in the table below. Construct the fractional flow curve for this reservoir. Capillary pressure gradients can be assumed negligible.

$$\phi = 23.5\%, S_{wi} = 25\%, \mu_w = 0.689 \text{ cp}, \mu_o = 1.97 \text{ cp}, B_o = 1.34 \text{ rb}, B_w = 1.248 \text{ rb}$$

STB

STB

$$k = 65 \text{ mD}, q_i = 1100 \text{ bbl/day}, \gamma_o = 0.85, \gamma_w = 1.015, A = 38000 \text{ ft}^2, \alpha = 30^\circ$$

S_{wi}	K_{ro}	K_{rw}
0.25	0.9142	0
0.35	0.8087	0.0216
0.45	0.6352	0.1094
0.5	0.4735	0.1562
0.6	0.1684	0.2352
0.7	0.0634	0.3476
0.82	0	0.4691

- b. Outline any five factors for consideration in water-flooding project
 ① Reservoir Geometry ② Fluid Properties ③ Reservoir Depth ④ Lithology & Rock P. ⑤ Fluid Saturation

- 5a. A water flood is under consideration for a narrow reservoir. It is proposed to drill a row of injection wells at one end of the reservoir. The reservoir is horizontal with the following data:

$$\text{Width} = 250 \text{ ft}, \text{length} = 1200 \text{ ft}, \text{height} = 45 \text{ ft}, \text{porosity} = 18\%, S_{wi} = 22\%, 1.89 \text{ cp}, \mu_w = 1 \text{ cp}, q_{inj} = 380 \text{ bbl/day}, S_{or} = 19.5\%, B_o = B_w = 1.$$

The saturation values are given in the table below

S_w	0.29	0.35	0.42	0.50	0.63	0.71	0.78

Using the following correlation to generate the relative permeability data.

$$K_{ro} = (1 - S_{WD})^3, K_{rw} = S_{WD}^3, S_{WD} = \frac{S_w - S_{wc}}{1 - S_{wc}}$$

- bi. What is water flooding for oil recovery
 ii. Define the term EOR process

$$F_w = \frac{1}{1 + \frac{K_{ro} - \mu_w - B_w}{K_{rw} \mu_o B_o}}, \quad \frac{K_{ro}}{K_{rw}} = a e^{-b S_w}$$

$$S_{or} = 1 - S_{wi}$$

$$S_{or} = 1 - S_{wi}$$

$$\frac{S_{or} - S_{wi}}{1 - S_{wi}}$$

$$S_{ur} = 1 - S_{or} - S_{wi}$$

$$0.22$$

SURE SUCCESS FOUNDATION

PNG 511: ENHANCED OIL RECOVERY (EOR) (2020/2021)

QUESTION 1

Given the permeability data in the question, the table can be completed to find $\left(\frac{k_{ro}}{k_{rw}}\right)$

S_w	0	10	20	30	40	50	60	70	75	80	90	100
k_{ro}	1.0	1.0	1.0	0.94	0.80	0.44	0.16	0.08	0	0	0	0
k_{rw}	0	0	0	0	0.04	0.11	0.20	0.30	0.36	0.44	0.68	1.0
$\frac{k_{ro}}{k_{rw}}$	∞	∞	∞	∞	1.82	4	0.8	0.27	0	0	0	0

A) To find the constant "a" and "b"; we use the given pivot data into the equation; $\frac{k_{ro}}{k_{rw}} = ae^{-bs_w}$

At 50% Water cut; $\frac{k_{ro}}{k_{rw}} = 4$

$$\therefore \frac{k_{ro}}{k_{rw}} = ae^{-bs_w}$$

$$4 = ae^{-0.5b} \dots \dots \dots (1)$$

Also, At 60% = 0.6 ; $\frac{K_{ro}}{K_{rw}} = 0.8$

$$\therefore \frac{K_{ro}}{K_{rw}} = ae^{-bS_w}$$

$$0.8 = ae^{-0.6b} \dots \dots \dots (2)$$

Dividing equation (i) by equation (ii)

$$\frac{4}{0.8} = \frac{ae^{-0.5b}}{ae^{-0.6b}}$$

$$5 = e^{-0.5b - (-0.6b)}$$

$$5 = e^{0.1b}$$

Taking the Natural log of both sides,

$$\ln(5) = 0.1b$$

$$\therefore b = \frac{\ln(5)}{0.1} = 16.094$$

Therefore, substitute (b) in equation (i).

$$4 = ae^{-0.5 \times 16.094}$$

$$4 = 3.2006 \times 10^{-4} a$$

$$a = \frac{4}{3.2006 \times 10^{-4}}$$

$$a = 12497.6307$$

$\therefore$ The value of the constants "a" and "b" are 16 and 12497.6307 respectively

B) Given the following parameters $\mu_o = 2.4 \text{ cp}$, $\mu_w = 0.68 \text{ cp}$,
 $\beta_o = 1.5 \text{ rb/STB}$ and $\beta_w = 1.05 \text{ rb/STB}$.

Therefore, the surface water cut with water saturation at 50% is;

$$f_w = \frac{1}{1 + \frac{k_{ro}}{k_{rw}} \frac{\mu_w \beta_w}{\mu_o \beta_o}}$$

$$\text{But } \frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$$

$$\therefore f_w = \frac{1}{1 + a e^{-b S_w} \frac{\mu_w \beta_w}{\mu_o \beta_o}}$$

$$f_w = \frac{1}{1 + 12497.6307 e^{-0.5 \times 16.094} \left[\frac{0.68 \times 1.05}{2.4 \times 1.5} \right]}$$

$$f_w = 0.5576$$

$$\approx 0.56$$

C) For the subsurface water cut (Reservoir)

$$f_w = \frac{1}{1 + \frac{k_{ro}}{k_{rw}} \frac{\beta_w \mu_w}{\beta_o \mu_o}}$$

$$\text{But } \frac{\beta_w}{\beta_o} \text{ is negligible, } \frac{k_{ro}}{k_{rw}} = a e^{-b S_w}$$

$$\therefore f_w = \frac{1}{1 + 12497.6 e^{-16.094 \times 0.5} \left(\frac{0.68}{2.4} \right)}$$

$$f_w = 0.4687 \approx 0.47$$

QUESTION 2

A) The ultimate recovery factor is different for different oil reservoirs because of their different reservoir characteristics and different recovery mechanisms.

For instance, the ultimate oil recovery factor of individual reservoir under primary and/or conventional recovery method may range from 5% of OOIP for the poorest reservoir characteristics to as high as 60% of OOIP possibly obtained from reservoir characteristics.

B) Producing reserves by energy source criteria can be defined as when the reserve are produced (depleted) using a natural energy inherent in the reservoir and/or through improved recovery method in addition to the primary reserve.

C) Polymer flooding is an Enhanced oil recovery technique used in the oil and gas industry to increase the production of oil from reservoirs. It involves injecting a solution of long-chain polymer molecules such as Xanthan gum into an oil reservoir to improve the mobility ratio and displace oil towards producing wells. This has been most often been achieved using high molecular weight water-soluble

Polymers of 2-Propanamide (acrylamide) and 2-propenoic acid (acrylic acid) as the partly neutralised sodium salt a ratio of about 70:30 of polymer to acid by weight. Polymer flooding will not only accelerate the oil recovery! can also increase it, in comparison to a normal water drive, because the by-passing of oil is greatly reduced.

Polymer injection involves injecting a water-soluble polymer mixed with water into the reservoir to improve the mobility of oil and reduce its viscosity.

D) i) Viscosity

ii) Capillary forces

iii) Gravity

iv) Capillary Pressure

v) Interfacial Tension

vi) Trapping in a single capillary: → Variation in radius
→ Variation in Interfacial Tension
→ Continuous Phases
→ Variation in Contact

E) For this solution;

REFER TO 2019/2020 solution - (Question 1)

QUESTION 3

Given the sandstone reservoir with relative permeability characteristics as shown in the table below;

S_w	0.24	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
k_{ro}	0.15	0.88	0.74	0.45	0.19	0.06	0.0	0.0	0.0
k_{rw}	0.0	0.01	0.04	0.09	0.17	0.28	0.44	0.67	1.00

and other fluid properties are given as: $\mu_o = 1.25 \text{ cP}$,
 $\mu_w = 0.76 \text{ cP}$, $\phi = 0.25$, $B_o = 1.126 \text{ bbl/STB}$, $\beta_w = 1.036 \text{ bbl/STB}$

A) Fractional flow of water (f_w) is given by the formula;

$$f_w = \frac{1 + \left(\frac{0.001127 k A}{\mu_o q_t} \right) \left[\frac{dP_c}{dx} - 0.483 \Delta P \sin(\alpha) \right]}{1 + \frac{k_{ro}}{k_{rw}} \frac{\mu_w}{\mu_o}}$$

$\therefore$ Since $\frac{\partial P_c}{\partial x} = 0$ and dip angle $\alpha = 0$; The formula is reduced to;

$$f_w = \frac{1}{1 + \frac{k_{ro}}{k_{rw}} \frac{\mu_w}{\mu_o}}$$

Next PAGE

∴ The new table with fractional flow (f_w) is generate below;

S_w	0.24	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.00
K_{ro}/K_w	∞	88	18.5	5	1.118	0.214	0	0	0
f_w	0	0.018	0.082	0.248	0.595	0.885	1	1	1

Plot the graph of f_w against corresponding S_w as seen in the graph attached below.

B) Surface KDR is given by the expression;

$$f_w = \frac{B_w KDR_s}{B_w KDR_s + B_o}$$

Solving for KDRs yields:

$$KDR_s = \frac{B_o}{B_w \left(\frac{1}{f_w} - 1 \right)} = \frac{B_o f_w}{B_w (1 - f_w)}$$

∴ The new table with Water-oil-Ratio (KDR) is generated below * with $B_o = 1.12 \text{ bbl/STB}$, $B_w = 1.03 \text{ bbl/STB}$.

S_w	0.24	0.30	0.40	0.50	0.60	0.70	0.80	0.90	1.0
f_w	0	0.018	0.082	0.248	0.595	0.885	1	1	1
KDR_s	0	0.02	0.10	0.359	1.60	8.368	∞	∞	∞

Plot the graph of WORs against corresponding S_w as seen in the graph attached below.

c) Using the Frontal Advance equation

$$(X)_{sw} = \left(\frac{5.615 i_w t}{\phi A} \right) \left(\frac{df_w}{dS_w} \right)_{sw}$$

where i_w = Water injection rate, bbl/day
 t = time, day.

$(X)_{sw}$ = Distance from the injection for any given Saturation S_w , ft

A = Area, ft²

$\left(\frac{df_w}{dS_w} \right)_{sw}$ = Position of displacement.

∴ Given the parameters;

$$(X)_{sw} = 500 \text{ ft}, \quad i_w = 0.427 \text{ bbl/day} = 0.427 \text{ bbl/day} \cdot$$

$$t = 100 \text{ hrs} = 4.167 \text{ days}, \quad A = 1 \text{ ft}^2.$$

$$\therefore 500 = \left(\frac{5.615 \times 0.427 \times 4.167}{0.25 \times 1} \right) \left(\frac{df_w}{dS_w} \right)_{sw}$$

$$500 = 39.9633 \left(\frac{df_w}{dS_w} \right)_{sw}$$

$$\left(\frac{df_w}{dS_w} \right)_{sw} = \frac{500}{39.9633} = 12.5115$$

D

D. recovery at breakthrough

$$X_p = \frac{\phi A L}{5.615 B_o} (\bar{S}_{wBT} - S_{wi})$$

From graph; $\bar{S}_{wBT} = 0.80$ (fw against graph)

$$X_p = \frac{0.25 \times 1 \times 500 (0.8 - 0.24)}{5.615 \times 1.12}$$

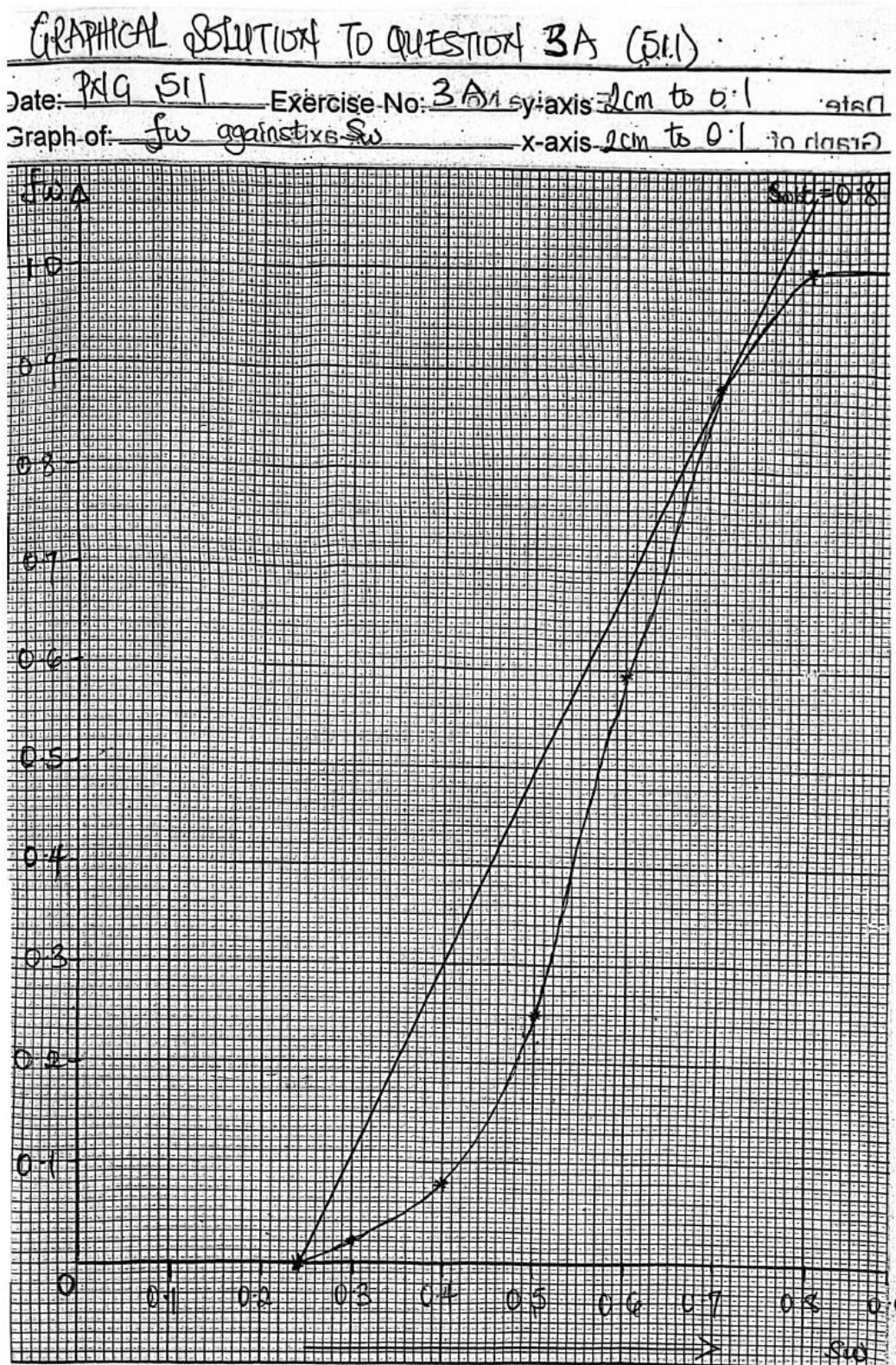
$$X_p = 11.13090 \text{ bbls.}$$

E.

$$f_w = \frac{k_{OR} R_w}{R_w k_{OR} + B_o}$$

$$= \frac{25000 (1.03)}{(1.03 \times 25000) + 1.12}$$

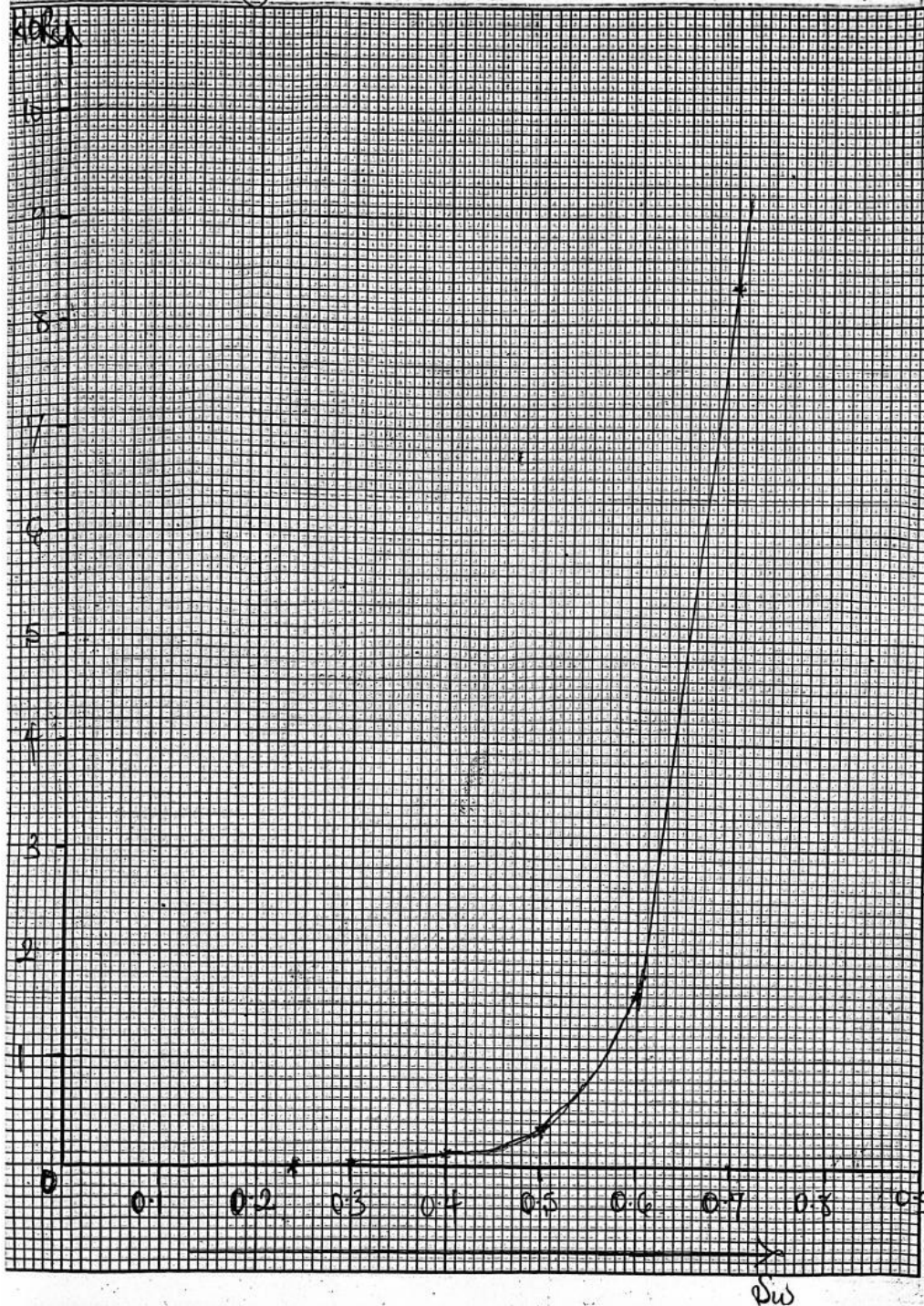
$$f_w = 0.99995 \approx 99.99\% \\ \approx 100\%$$



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GRAPHICAL SOLUTION TO QUESTION 3B (511).

ate: PKG 511 Exercise No: 3B y-axis: 2m to 1 Date: _____
 graph of: HRS against Sw x-axis: " " 0.1



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QUESTION FOUR (4)

Given the following parameters; $\phi = 23.5\%$, $S_{wi} = 25\%$,
 $\mu_w = 0.689 \text{ cP}$, $\mu_o = 1.97 \text{ cP}$, $\beta_o = 1.34 \text{ rb/STB}$, $\beta_w = 1.248 \frac{\text{rb}}{\text{STB}}$
 $K = 65 \text{ mD}$, $q_t = 1100 \text{ bbl/day}$, $\gamma_o = 0.85$, $\gamma_w = 1.015$
 $A = 38000 \text{ ft}^2$ and $\alpha = 30^\circ$.

Capillary pressure gradient is negligible, therefore;

$$\frac{dp_c}{dx} = 0$$

The fractional flow (f_w) can be calculated using the formula;

$$f_w = \frac{1 + \left(\frac{0.001127 K K_o A}{\mu_o q_t} \right) \left[\frac{dp_c}{dx} - 0.433 \Delta \rho \sin(\alpha) \right]}{1 + \frac{K_{ro}}{K_{rw}} \frac{\mu_w}{\mu_o} \frac{\beta_w}{\beta_o}}$$

However, since $\frac{dp_c}{dx} = 0$, and for the reservoir f_w , $\frac{\beta_w}{\beta_o}$ is negligible ($\frac{\beta_w}{\beta_o}$).

Therefore the formula becomes;

$$f_w = \frac{1 - \left(\frac{0.001127 K K_o A}{\mu_o q_t} \right) (0.433 \Delta \rho \sin(\alpha))}{1 + \frac{K_{ro}}{K_{rw}} \frac{\mu_w}{\mu_o}}$$

But $\Delta \rho = \text{Water-oil density difference, g/cm}^3$

$$\Delta \rho = \frac{\rho_w - \rho_o}{62.4} \text{ (g/cm}^3\text{)}.$$

$$\therefore l_w; \gamma_w = \frac{l_w}{62.4} \quad l_w = \gamma_w \times 62.4 \\ = 1.015 \times 62.4 = 63.336$$

$$l_o = \gamma_o \times 62.4 = 0.85 \times 62.4 = 53.04$$

$$\therefore \Delta \rho = \frac{63.336 - 53.04}{62.4} = 0.165 \text{ g/cm}^3$$

or simply;

$$\Delta \rho = \gamma_w - \gamma_o = 1.015 - 0.85 = 0.165 \text{ g/cm}^3$$

(Note; first approach is used when densities is given in lb, however given $\rho \cdot a$, you can subtract directly).

$$\therefore \text{for } S_{wc} = 0.25, \frac{K_{ro}}{K_{rw}} = \frac{0.9142}{0} = \infty$$

$$f_w = \frac{1 - \left(\frac{0.001127 \times 60 \times 0.9142 \times 38000}{1.97 \times 1100} \right) (0.433 (0.165))}{1 + \infty}$$

$$f_w = 0$$

$$\text{for } S_w = 0.35, \frac{K_{ro}}{K_{rw}} = \frac{0.8087}{0.0216} = 37.4398$$

$$f_w = \frac{1 - \left(\frac{0.001127 \times 0.8087 \times 60 \times 38000}{1.97 \times 1100} \right) (0.433 \times 0.165 \sin(36))}{1 + 37.4398 \times \left(\frac{0.689}{1.97} \right)}$$

$$f_w = 0.0685$$

The remaining Saturation values are completed with the table generated as;

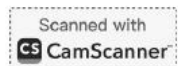
S_w	K_{ro}/K_{rw}	f_w
0.25	∞	0
0.35	37.4398	0.0685
0.45	5.8062	0.3211
0.50	3.0314	0.4756
0.6	0.7160	0.7940
0.7	0.1824	0.9375
0.82	0	1

Plot the graph of f_w against corresponding S_w as see in the graph attached below to develop the fractional flow curve

B) Factors to be considered for water-flooding Projects;

⇒ Reservoir geometry-

- ⇒ Fluid properties
- ⇒ Reservoir depth
- ⇒ Lithology and rock properties
- ⇒ Fluid saturations



Paper 2 of 2 — PEE 511 (Secondary & Enhanced Oil Recovery)

Level 500 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



COLLEGE OF TECHNOLOGY
DEPARTMENT OF PETROLEUM AND NATURAL GAS ENGINEERING

B.Sc. DEGREE EXAMINATION 2018/2019 FIRST SEMESTER EXAMINATION APRIL, 2019

COURSE: PNG 511 **Secondary & Enhanced Oil Recovery Principles** **TIME: 3 Hrs**

Instruction: Answer any four questions

QUESTION ONE

A waterflood is under consideration for a narrow "shoestring" reservoir that is 300ft wide, 20 ft thick and 1000 ft long. The reservoir is horizontal and has a porosity of 0.15 and an initial water saturation of 0.363 which is considered immobile. It is proposed to drill a new row of injection well at one end of the reservoir and flood the reservoir by injecting water at a rate of 338 B/D. viscosity of the oil and water are 2.0 and 1.0 cp respectively. Relative permeability data corresponding to the displacement of oil by water are given by the two equations below. The residual oil saturation is 0.205. base permeability is the absolute permeability to oil at interstitial water saturation, which is assumed to be equal to the absolute permeability. Oil and water FVF's are 1.0

and
$$K_{ro} = (1 - S_{WD})^{2.56}$$

where
$$K_{rw} = 0.78 S_{WD}^{3.72}$$

$$S_{WD} = \frac{(S_w - S_{wi})}{(1 - S_{or} - S_{wi})}$$

Which represent normalized water saturation. Estimate oil recovery at breakthrough, time to breakthrough and WOR at breakthrough using incremental saturation interval of 0.05.

QUESTION TWO

a) Write short note on the following:

- Primary oil recovery
- Secondary oil recovery
- Tertiary oil recovery
- Enhanced oil recovery

b) A waterflood is to be conducted in an undersaturated oil reservoir which has dimensions that will result in linear flow. The average cross-sectional area is approximately 78,000 square feet. Additional reservoir data are

$i_w = 7000$ bbl/day	$B_w = 1.02$ RB/STB	$k = 50$ md
$S_{wi} = 25\%$	$\mu_o = 1.39$ cp	$B_o = 1.25$ RB/STB
$\phi = 22\%$	$\mu_w = 0.50$ cp	$\gamma = 0$

1

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S_w	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.72
k_{ro} / k_{rw}	∞	36.95	11.12	4.84	2.597	1.340	0.612	0.292	0.098	0.017	0.00

If the first row of producers is located 1320 feet from the injection wells.

- Determine the oil recovery (STB) at the time of breakthrough.
- Determine the time until breakthrough, days.
- Determine the displacement sweep efficiency at the time of breakthrough.
- How many barrels of water must be injected to obtain breakthrough?

QUESTION THREE

The following data are available for a linear-reservoir system:

S_w	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75
k_{ro}/k_{rw}	30.23	17.0	9.56	5.38	3.02	1.70	0.96	0.54	0.30	0.17	0.10

Oil formation volume factor $B_o = 1.25$ bbl/STB
 Water formation volume factor $B_w = 1.02$ bbl/STB
 Formation thickness $h = 20$ ft
 Cross-sectional area $A = 26,400$ ft²
 Porosity $\phi = 25\%$
 Injection rate $i_w = 900$ bbl/day
 Distance between producer and injector $L = 600$ ft
 Oil viscosity $\mu_o = 2.0$ cp
 Water viscosity $\mu_w = 1.0$ cp
 Dip angle $\gamma = 0^\circ$
 Connate water saturation $S_{wc} = 20\%$
 Initial water saturation $S_{wi} = 20\%$
 Residual water saturation $S_{or} = 20\%$

- Calculate the water saturation profile after 60, 120, and 240 days.
- What is the optimum Time to breakthrough?

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QUESTION FOUR

The Dykstra and Parsons (1950) permeability ordering approach is used to describe a reservoir by the following five-layer system:

Layer	k, md	h, ft
1	120	15
2	90	15
3	70	15
4	55	10
5	30	10

The reservoir is under consideration for further development by water injection. The following additional information is available:

$$k_{rw@S_{or}} = 0.3$$

$$k_{ro@S_{wi}} = 0.9$$

$$\mu_o = 2.0 \text{ cp}$$

$$\mu_w = 0.5 \text{ cp}$$

$$B_o = 1.20 \text{ bbl/STB}$$

$$B_w = 1.01 \text{ bbl/STB}$$

$$h_t = 60 \text{ ft}$$

Calculate the vertical sweep efficiency and surface water-oil ratio using Stiles' method.

QUESTION FIVE

(a) The following data are available for a reservoir system:

S_w	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70	0.75
f_w	0.062	0.105	0.173	0.271	0.398	0.541	0.677	0.788	0.869	0.922	0.956

$$\mu_o = 2.0 \text{ cp}$$

$$B_o = 1.25 \text{ bbl/STB}$$

$$\phi = 25\%$$

$$S_{wi} = 20\%$$

$$I_w = 900 \text{ bbl/day}$$

$$N_s = 496,449 \text{ STB}$$

$$EV = 100\%$$

$$\mu_w = 1.0 \text{ cp}$$

$$B_w = 1.02 \text{ bbl/STB}$$

$$h = 20 \text{ ft}$$

$$S_{or} = 20\%$$

$$(PV) = 775,779 \text{ bbl}$$

$$EA = 100\%$$

Predict the water flood performance to abandonment at a WORs of 45 STB/STB.

(b) Draw and explain briefly the following regular injection patterns

- Direct line drive.
- Staggered line drive
- Five spot.
- Seven spot
- Nine spot

3

$$E_d = \frac{\bar{S}_{wt} - S_{wi}}{1 - S_{wi}}$$

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QUESTION ONE.

Given,

$$\text{width} = 300\text{ft}$$

$$\text{thickness} = 20\text{ft}$$

$$\text{length} = 1000\text{ft}$$

$$\phi = 0.15$$

$$S_{wi} = 0.363$$

$$i_w = 338\text{B/D}$$

$$\mu_o = 2.0\text{cp}$$

$$\mu_w = 1.0\text{cp}$$

$$S_{or} = 0.205$$

$$B_w = 1.0$$

$$B_o = 1.0$$

$$\text{Incremental Saturation Interval} = 0.05$$

$$k_{ro} = (1 - S_{wo})^{2.56}$$

$$k_{rw} = 0.78 S_{wo}^{3.72}$$

$$S_{wo} = \frac{(S_w - S_{wi})}{(1 - S_{or} - S_{wi})}$$

Solution.

$$1 - S_{or} = 1 - 0.205 \\ = 0.795$$

Tabulate your values for the various values of saturations for k_{ro} , k_{rw} , S_{wo} using the formula given above.

(1)

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S/N	S_w	S_{wD}	K_{rw}	K_{ro}	$f_w = \left(\frac{1}{1 + \frac{K_{rw}}{K_{ro}}} \right)$
1	0.363	0	0	1	7.0011×10^{-4}
2	0.413	0.1157	2.557×10^{-4}	0.72995	0.013072
3	0.463	0.2315	3.375×10^{-3}	0.50962	0.08331
4	0.513	0.34722	0.01525	0.33559	0.3042
5	0.563	0.4630	0.0445	0.2036	0.6509
6	0.613	0.5787	0.10196	0.10938	0.8931
7	0.663	0.6944	0.20086	0.0481	0.9805
8	0.713	0.8102	0.35650	0.0142	0.99890
9	0.763	0.9259	0.5858	1.2786×10^{-3}	

$$S_{wD} = \frac{(S_w - S_{wi})}{(1 - S_{or} - S_{wi})}$$

$$= \frac{0.363 - 0.363}{1 - 0.205 - 0.363}$$

$$= 0$$

$$K_{rw} = 0.78 S_{wD}^{3.72}$$

$$= 0.78 (0)^{3.72}$$

$$= 0$$

$$K_{ro} = (1 - S_{wD})^{2.56}$$

$$= (1 - 0)^{2.56}$$

$$= 1$$

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Estimate Oil Recovery at breakthrough, time to breakthrough and WOR at breakthrough.

② Oil Recovery at breakthrough.

$$N_p = \frac{QAL}{S \cdot 615 B_o} (\bar{S}_{wt} - S_{wi}) \quad \text{from graph } \bar{S}_{wt} = 0.70.$$

$$= \frac{0.15 \times 300 \times 20 \times 1000}{S \cdot 615 \times 1} \times (0.70 - 0.363)$$

$$= 160284.95 \times (0.337)$$

$$= \underline{\underline{54016.03 \text{ stb.}}}$$

③ Time to breakthrough

$$t = \frac{PV Q_{IBT}}{i_w}$$

$$Q_{IBT} = (\bar{S}_{wt} - S_{wi}) = (0.70 - 0.363) = 0.337.$$

$$PV = \frac{QAL}{S \cdot 615} = \frac{0.15 \times 300 \times 20 \times 1000}{S \cdot 615} = 160284.95$$

$$t = \frac{PV Q_{IBT}}{i_w}$$

$$= \frac{160284.95 \times 0.337}{338}$$

$$= 159.8 \approx \underline{\underline{160 \text{ days.}}}$$

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© WOR at breakthrough.

$$WOR_r = \frac{f_w}{1 - f_w} \quad \text{at Reservoir Condition}$$

from graph, $f_w = 0.891$.

$$WOR_r = \frac{0.891}{1 - 0.891} = \underline{\underline{8.174 \text{ bbl/bbl}}}$$

WOR at surface;

$$WOR_s = \frac{B_o WOR_r}{B_w}$$

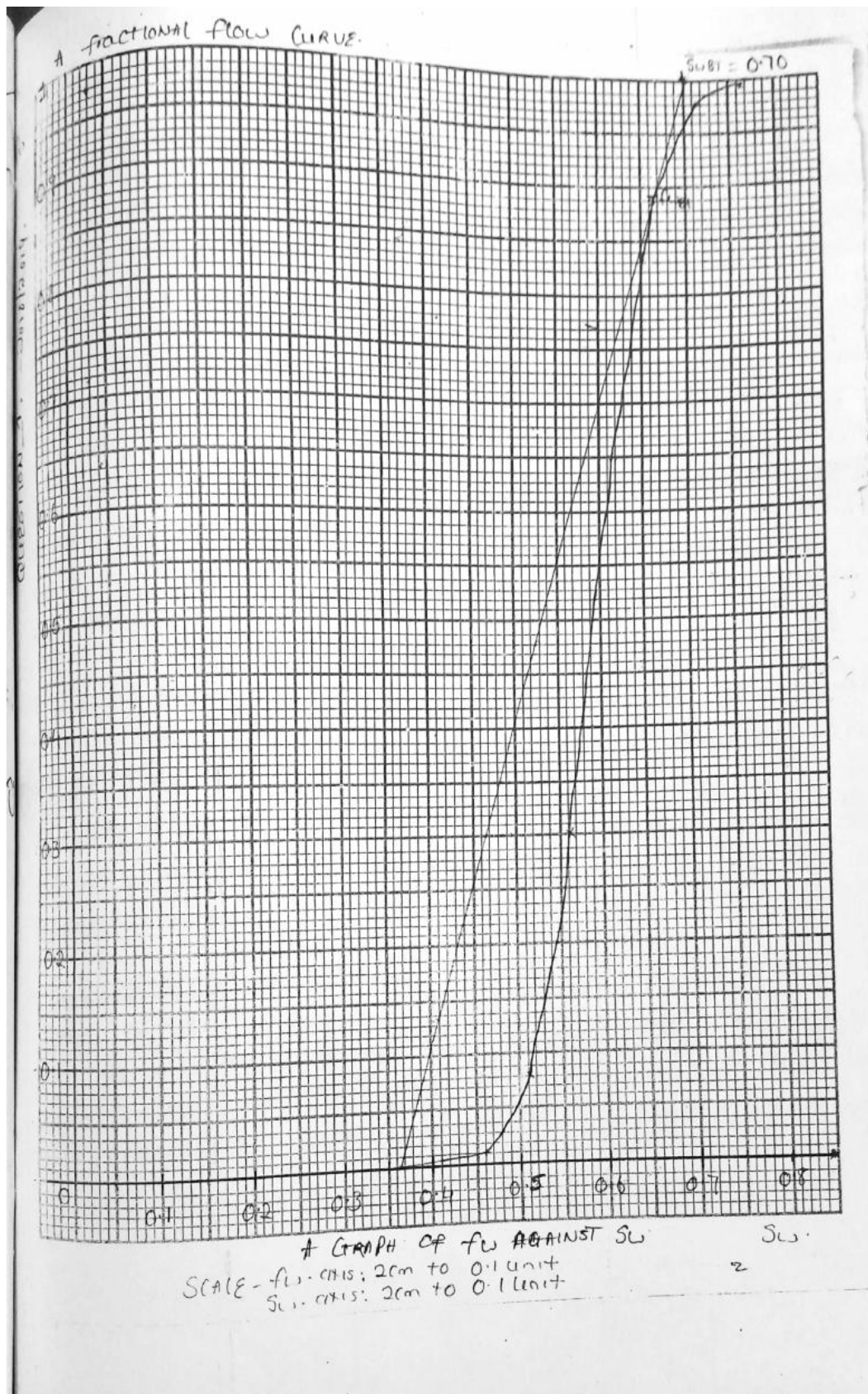
$$= \frac{1 \times 8.174}{1}$$

$$= \underline{\underline{8.174 \text{ stb/stb}}}$$

Since it is WOR at breakthrough, means the WOR when water is been produced at the surface

$$WOR_s = \underline{\underline{8.174 \text{ stb/stb}}}$$

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Question Two

Q) Write short note on the following:

1) Primary Oil Recovery - is the process of Recovery (production) of Hydrocarbon from a reservoir by allowing the oil to flow naturally to the bottom of the well due to gravity and difference in pressure (pressure gradient between the Reservoir and the bottom of the well).

2) Secondary Oil Recovery - Secondary Recovery refers to process of producing H/C from the reservoir by increasing the natural energy of the reservoir usually by injecting some fluid into the Reservoir in order to maintain or boost the reservoir pressure and thereby pushing the oil towards the producing well. Includes injection of Gas or water.

3) Tertiary Oil Recovery:

This is the process of Recovering of the remaining oil in the Reservoir after a Conventional Primary and Secondary Project (phases) such as water flooding. Tertiary Recovery is any technique applied after a Secondary recovery.

4) Enhanced Oil Recovery:

This entails changing the actual properties of the Hydrocarbon by altering the relative oil permeability characteristic of the Hydrocarbon which further distinguishes this phase of Recovery from the Secondary Recovery method.

Graph 2

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2b.

$$A = 78,000 \text{ square feet}$$

$$i_w = 7000 \text{ bbl/day}$$

$$B_w = 1.02 \text{ RB/stb}$$

$$k = 50 \text{ md}$$

$$S_{wi} = 25\%$$

$$\mu_o = 1.39 \text{ cp}$$

$$B_o = 1.25 \text{ RB/stb}$$

$$\phi = 22\%$$

$$\mu_w = 0.5 \text{ cp}$$

$$r = 0.$$

S_w	$\frac{k_{rg}}{k_{rw}}$	$f_w = \frac{1}{1 + \frac{k_{ro} \mu_o}{k_{rw} \mu_w}}$
0.25	∞	0.000
0.30	36.95	0.070
0.35	11.12	0.200
0.40	4.84	0.365
0.45	2.597	0.547
0.50	1.340	0.674
0.55	0.612	0.820
0.60	0.292	0.905
0.65	0.098	0.966
0.70	0.017	0.994
0.72	0.00	1.000

Areal and Vertical Sweep Efficiencies are assumed to be further, The Capillary Pressure gradient can be neglected. Neglecting gravity and capillary forces, the fractional flow equation reduce to the for,

$$f_w = \frac{1}{1 + \frac{k_{ro} \mu_o}{k_{rw} \mu_w}}$$

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If the first row of producers is located 1320 feet from the injection wells.

(a) Determine the Oil Recovery (Colb) at the time of breakthrough.

$$N_p = \frac{QAL}{S_{wi} B_o} (\bar{S}_{wt} - S_{wi})$$

from graph $\bar{S}_{wt} = 0.615$

$$= \frac{0.22 \times 78,000 \times 1320 (0.615 - 0.25)}{5.615 \times 1.25}$$

$$= 1,177,943.08 \text{ stb.}$$

(b) Determine the time until breakthrough.

$$t_{bt} = \frac{P_v \times Q_{ibt}}{i_w}$$

$$Q_{ibt} = (\bar{S}_{wt} - S_{wi}) = (0.615 - 0.25) = 0.365$$

$$t_{bt} = \frac{QAL}{S_{wi}} + \frac{Q_{ibt}}{i_w}$$

$$= \frac{0.22 \times 78,000 \times 1320 \times 0.365}{5.615 + 7000}$$

$$= 210 \text{ days}$$

(c) Determine the displacement sweep efficiency at the time of breakthrough

$$E_D = \frac{\bar{S}_{wt} - S_{wi}}{1 - S_{wi}} = \frac{0.615 - 0.25}{1 - 0.25}$$

$$= 0.487$$

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⑧ How many barrels of water must be injected to obtain breakthrough?

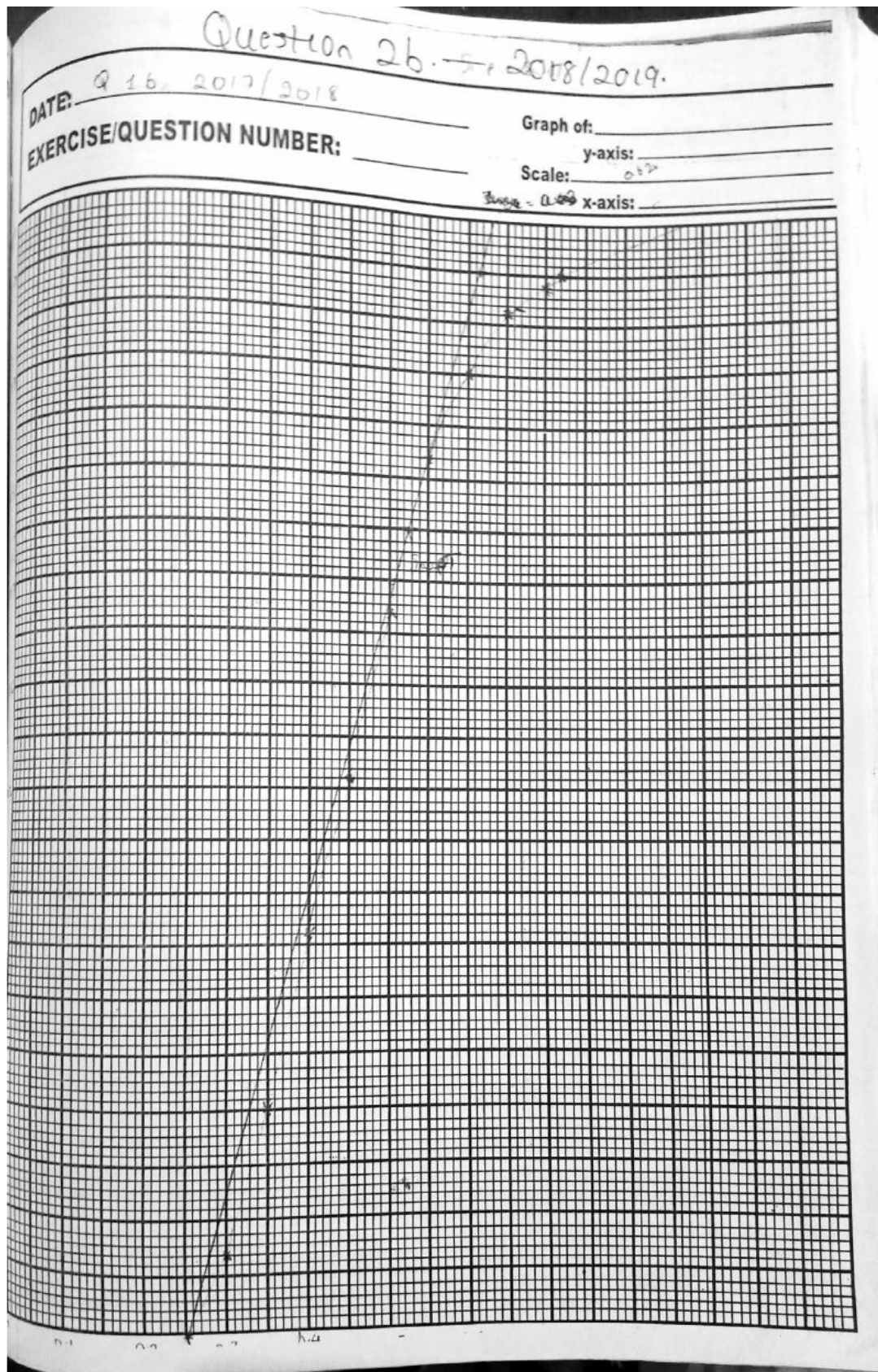
$$W = P V Q_{ibt}$$

$$= \frac{QAL}{5.615} \times 0.365$$

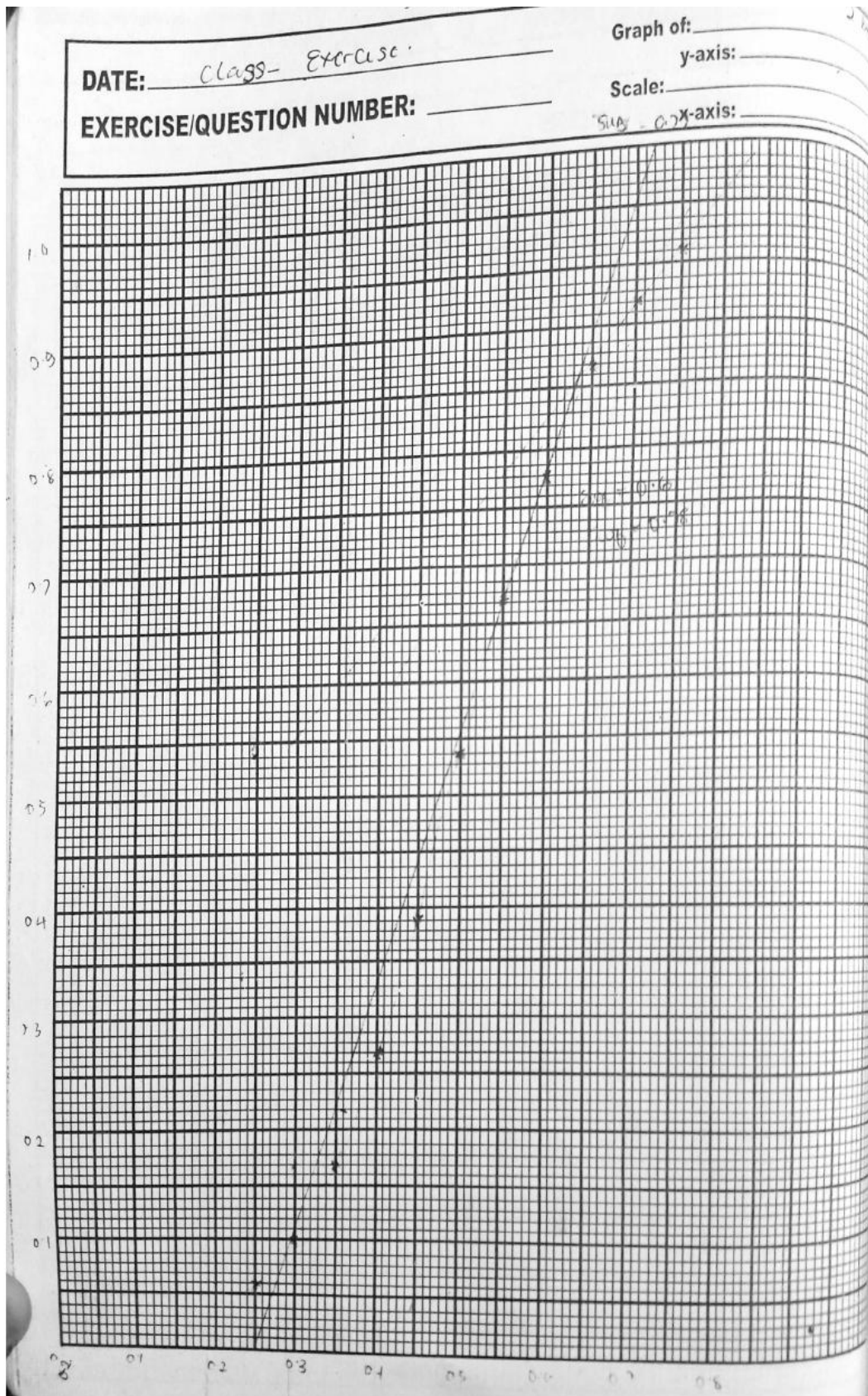
$$= \frac{0.22 \times 18,000 \times 1320}{5.615} \times 0.365$$

$$= \underline{\underline{1,472,428.8 \text{ bbls}}}$$

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QUESTION THREE

	k_{ro}/k_{rw}	$f_w = \frac{1}{1 + \frac{k_{ro} \mu_w}{k_{rw} \mu_o}}$
0.25	30.23	0.062
0.30	17.0	0.105
0.35	9.56	0.173
0.40	5.38	0.271
0.45	3.02	0.398
0.50	1.70	0.541
0.55	0.96	0.677
0.60	0.54	0.788
0.65	0.30	0.869
0.70	0.17	0.922
0.75	0.10	0.956

$B_o = 1.25 \text{ bbl/stb.}$
 $B_w = 1.02 \text{ bbl/stb.}$
 $h = 20 \text{ ft}$
 $A = 26400 \text{ ft}^2$
 $\sigma = 25\%$
 $i_w = 900 \text{ bbl/day}$
 $L = 600 \text{ ft}$
 $\mu_o = 2.0 \text{ cp}$

$\mu_w = 1.0 \text{ cp}, \gamma = 0^\circ$
 $S_{wc} = 20\%$
 $S_{wi} = 20\%$
 $S_{or} = 20\%$

$$f_w = \frac{1}{1 + 30.23 \times \frac{1}{2}} = \underline{\underline{0.06205}}$$

@ Calculate the water saturation profile after 60, 120, and 240 days.

Given 3

6

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from fractional flow curve.

$$\bar{S}_{wb} = 0.710.$$

$$Q_{wf} = \bar{S}_w - S_{wi}$$

$$= 0.710 - 0.2.$$

$$Q_{wf} = 0.510.$$

$$Q = \frac{1}{\text{slope.}}$$

$$= \frac{1}{0.510} = \underline{\underline{1.9608}}$$

from graph.

$$S_{wf} = 0.596.$$

$$f_{wf} = 0.77 \approx \underline{\underline{77\%}}$$

$$X_s = \frac{5.615 \text{ iwt}}{O_A} \left(\frac{\partial f_w}{\partial S_w} \right).$$

$$= \frac{5.615 \times 900 \times t}{0.25 \times 26400} \left(\frac{\partial f_w}{\partial S_w} \right)$$

$$t_s = 0.77 t \left(\frac{\partial f}{\partial S_w} \right)$$

So for the saturation table, substitute the other value of slope and time each.

N.B - To get the other values of slope for the various assumed saturation from $S_{wf}(0.6, 0.65, 0.7, 0.75)$ either trace from the S_{wf} axis to the curve and draw a tangent then get slope or use the formula method.

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$$\left(\frac{\partial f_w}{\partial s_w}\right)_{s_w} = -\left(\frac{\mu_w}{\mu_o}\right) a b e^{b s_w}$$

$$\left(1 + \left(\frac{\mu_w}{\mu_o}\right) a b e^{b s_w}\right)^2$$

for further understanding

Check Question five Graphical Solution and Slope Estimation

So substitute the value of μ_w, μ_o, a, b at the various f_w

To get a and b

plot k_o/k_w against saturation on a semi-log graph and determine a and b constant

But for examination condition use the graphical method to get saturation thereby getting the slope $\left(\frac{\partial f_w}{\partial s_w}\right)$ at each saturation

Assumed s_w	$\partial f_w / \partial s_w$	$x = 0.77 + \frac{\partial f_w}{\partial s_w}$ @ 60 days	$x = 0.77 + \frac{\partial f_w}{\partial s_w}$ @ 120 days	$x = 0.77 + \frac{\partial f_w}{\partial s_w}$ @ 240 days
0.596	1.960	90.552	180.084	358.68
0.6	1.923	88.7964	176.59	351.726
0.65	1.31	60.6606	120.63	240.279
0.70	0.8312	38.4014	76.37	152.1096
0.75	0.5012	23.15544	46.050	91.7196

6) What is the optimum time to breakthrough.

$$P_v = \frac{\phi A L}{5.615} = \frac{0.25 \times 26,400 \times 660}{5.615} = 775,779.666$$

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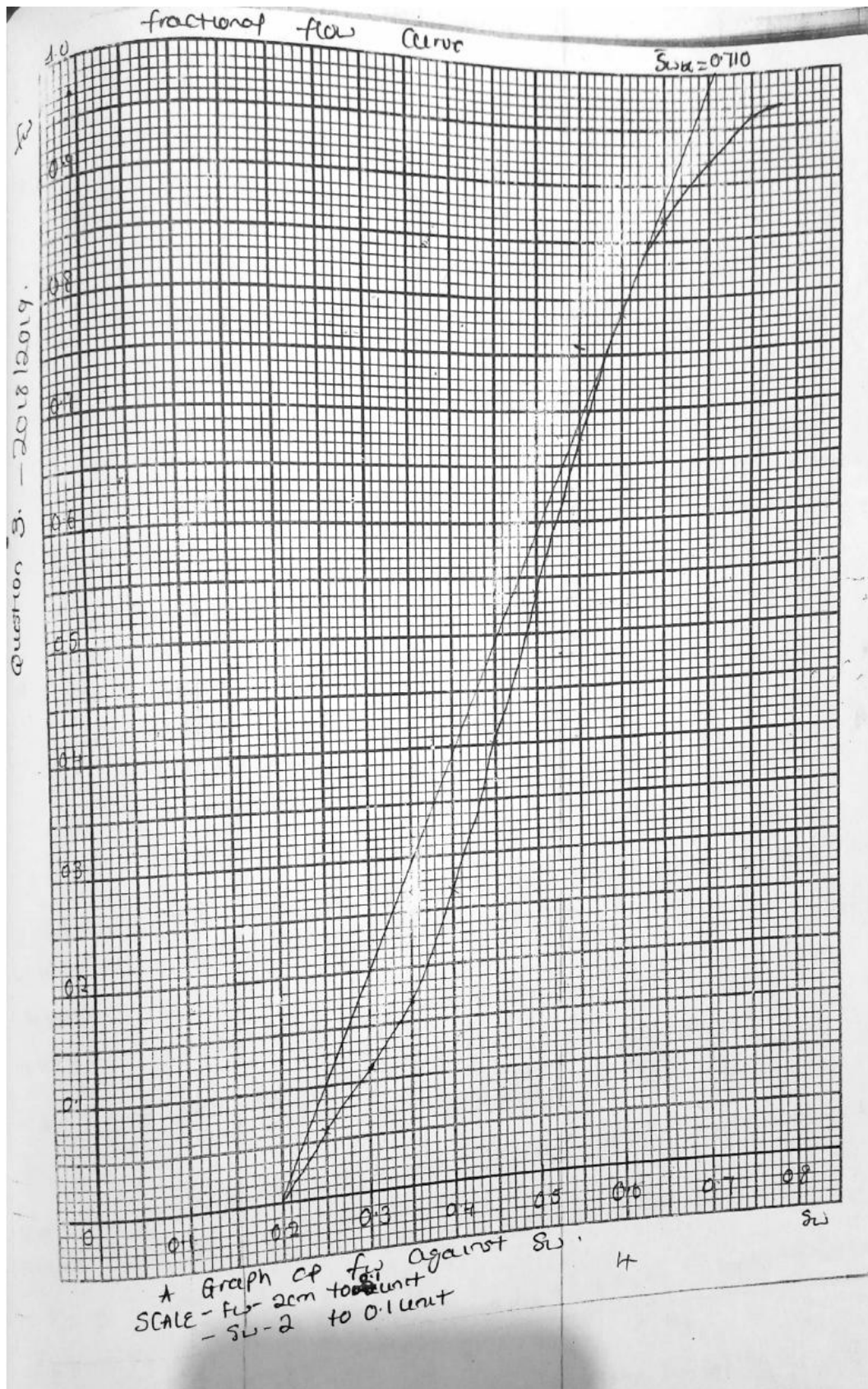
*to Calculate time to break-through

$$t_{BT} = \frac{P_0}{i_w} + \frac{1}{\left(\frac{df_w}{ds_w}\right)_{sw}}$$

$$= \frac{776,779}{900} \times \left(\frac{1}{1.960}\right)$$

$$= 439.78 \text{ days} \approx \underline{\underline{440 \text{ days}}}$$

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Question four.

layer.	K_{md}	h, ft
1	120	15
2	90	15
3	70	10
4	55	10
5	30	10

The Reservoir is under consideration for further development by water injection. The following additional information is available:

$$K_{rw} \odot S_{or} = 0.3$$

$$K_{ro} \odot S_{wi} = 0.9$$

$$\mu_o = 2.0 \text{ cp}$$

$$\mu_w = 0.5 \text{ cp}$$

$$B_o = 1.20 \text{ bbl/stb.}$$

$$B_w = 1.01 \text{ bbl/stb.}$$

$$h_t = 60 \text{ ft}$$

Calculate the vertical sweep efficiency and surface water-oil ratio using Stiles' method.

Solution.

Step 1. - Calculate parameter A

$$A = \frac{K_{rw}}{K_{ro}} \frac{\mu_o}{\mu_w} \frac{B_o}{B_w} = \frac{0.3 \times 2.0 \times 1.2}{0.9 \times 0.5 \times 1.01} = 1.584 \text{ ft}^2$$

Step 2. - Calculate E_v and WORs when breakthrough occurs in the first layer.

$$* i = 1 \text{ layer 1}$$

$$E_v = \frac{k_i \sum_{j=1}^i h_j + \sum_{j=2}^5 E_{kh} j}{k_i h_t} = \frac{k_i h_1 + (k_2 h_2 + k_3 h_3 + k_4 h_4 + k_5 h_5)}{k_i h_t}$$

$$= \frac{120 \times 15 + (90 \times 15 + 70 \times 10 + 55 \times 10 + 30 \times 10)}{120 \times 60} = 0.653$$

Gresly

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$$\begin{aligned}
 WOR_5 &= 1.584 \frac{\sum_{j=1}^5 (kh)_j}{\sum_{j=2}^5 (kh)_j} \\
 &= \frac{1.584 \times (kh)_1}{(kh)_2 + (kh)_3 + (kh)_4 + (kh)_5} \\
 &= \frac{1.584 \times 120 \times 15}{(90 \times 15) + (70 \times 10) + (55 \times 10) + (30 \times 10)} \\
 &= \underline{\underline{0.983 \text{ stb/sth}}}.
 \end{aligned}$$

$$\begin{aligned}
 *i=2. \\
 E_U &= \frac{k_2 \sum_{j=2}^2 h_j \sum_{j=3}^5 (kh)_j}{k_2 h_1} = \frac{90 (h_1 + h_2) + (k_3 h_3 + k_4 h_4)}{90 \times 60} \\
 &= \frac{90 (15 + 15) + (70 \times 10 + 55 \times 10 + 30 \times 10)}{90 \times 60} \\
 &= \underline{\underline{0.787}}
 \end{aligned}$$

$$\begin{aligned}
 WOR_5 &= 1.584 \frac{\sum_{j=2}^2 (kh)_j}{\sum_{j=3}^5 (kh)_j} = 1.584 \frac{(kh)_1 + (kh)_2}{(70 \times 10 + 55 \times 10 + 30 \times 10)} \\
 &= \frac{1.584 [(120 \times 15) + (90 \times 15)]}{(70 \times 10 + 55 \times 10 + 30 \times 10)} \\
 &= \underline{\underline{3.22 \text{ stb/sth}}}.
 \end{aligned}$$

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$$* i=3.$$

$$E_v = \frac{k_3 \sum_{j=3}^3 h_j + \sum_{j=4}^5 (kh)_j}{k_3 h_t}$$

$$= \frac{70 (15+15+10) + (55 \times 10 + 30 \times 10)}{70 \times 60}$$

$$= \underline{\underline{0.8690}}$$

$$WORS = \frac{1.584 \sum_{j=3}^3 (kh)_j}{\sum_{j=4}^5 (kh)_j} = \frac{1.584 (120 \times 15 + 90 \times 15 + 70 \times 10)}{850}$$

$$= \frac{6098.4}{850} = \underline{\underline{7.175}}$$

$$* i=4.$$

$$E_v = \frac{k_4 \sum_{j=4}^4 h_j + \sum_{j=5}^5 (kh)_j}{k_4 h_t}$$

$$= \frac{55 (150+15+10+10) + (30 \times 10)}{55 \times 60}$$

$$= \frac{2,750 + 300}{3,300}$$

$$= \underline{\underline{0.9242}}$$

$$WORS = \frac{1.584 \sum_{j=4}^4 (kh)_j}{\sum_{j=5}^5 (kh)_j} = \frac{1.584 (120 \times 15 + 90 \times 15 + 70 \times 10 + 55 \times 10)}{300}$$

$$= \frac{6,969.6}{300}$$

$$= \underline{\underline{23.232}}$$

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$$* i = 5$$

$$E_v = \frac{k_5 \sum_{j=5}^5 h_j + \sum_{j=6}^5 (kh)_j}{k_5 h_e}$$

$$= \frac{30 (15+15 + 10+10+10) + 0}{30 \times 60}$$

$$= \frac{1800}{1800} = 1$$

$$WOR = 1.584 \frac{\sum_{j=5}^5 (kh)_j}{\sum_{j=6}^5 (kh)_j}$$

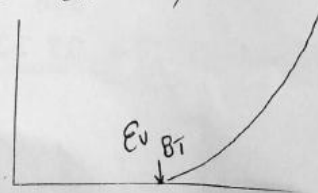
$$= 1.584 \times \frac{120 \times 15 + 90 \times 15 + 70 \times 10 + 55 \times 10}{0}$$

$$= \frac{1.584 \times 4700}{0}$$

$$= \frac{7444.8}{0} = -$$

n	E_v	WOR_s
1	0.653	0.983
2	0.787	3.22
3	0.8690	7.175
4	0.9242	23.232
5	1.00	

The graph shows the relationship between the Vertical Sweep efficiency and Producing WOR.



On a Cartesian

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LIVE DITION +102.

Given

$$\mu_o = 2.0 \text{ cP}$$

$$B_o = 1.25 \text{ bbl/stb.}$$

$$\phi = 25\%$$

$$S_{wi} = 20\%$$

$$i_w = 900 \text{ bbl/day}$$

$$N_s = 4961449 \text{ stb.}$$

$$E_v = 100\%$$

$$\mu_w = 1.0 \text{ cP}$$

$$B_w = 1.02 \text{ bbl/stb.}$$

$$h = 20 \text{ ft}$$

$$S_{or} = 20\%$$

$$P_v = 775,779 \text{ bbl.}$$

$$E_A = 100\%$$

S_w	f_w
0.25	0.062
0.30	0.106
0.35	0.173
0.40	0.271
0.45	0.398
0.50	0.541
0.55	0.677
0.60	0.788
0.65	0.869
0.70	0.922
0.75	0.956

Predict the water flood performance to abandonment at a WORs of 45 stb/stb.

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Solution.

Plot f_w vs S_w and construct the tangent to the curve. Extrapolate the tangent to $f_w = 1.0$ and determine:

$$S_{wf} = S_{wBT} = 0.596$$

$$f_{wf} = f_{wBT} = 0.780$$

$$Q_{iBT} = (S_{wBT} - S_{wi}) = (0.707 - 0.2) = 0.507$$

$$\text{Slope} = \frac{1}{Q_{iBT}} = \frac{1}{0.507} = 1.97$$

• Calculating E_{oBT} Displacement Efficiency.

$$\begin{aligned} E_{oBT} &= \frac{S_{wBT} - S_{wi}}{1 - S_{wi}} \\ &= \frac{0.707 - 0.2}{1 - 0.2} \\ &= \underline{0.634} \end{aligned}$$

• Calculating $(N_p)_{BT}$

$$\begin{aligned} (N_p)_{BT} &= N_s E_D \\ &= 496,499 \times 0.634 \\ &= \underline{314,780 \text{ STB}} \end{aligned}$$

• Calculate Cumulative water injected at breakthrough

$$\begin{aligned} W_{iBT} &= P V Q_{iBT} \\ &= 775,779 \times 0.507 \\ &= \underline{393,198 \text{ bbl}} \end{aligned}$$

• Calculate the time to breakthrough.

$$t_{BT} = \frac{W_{iBT}}{i_w} = \frac{393,198}{900} = 436.88 \text{ days}$$

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Assuming other values of saturations, 0.60, 0.70, 0.80.
 from graph, trace the saturation to where it intersects the curve, draw a tangent at that point as in the graph,
 then calculate slope.

$$\frac{\partial f_w}{\partial S_w} = \frac{\Delta f_w}{\Delta S_w}$$

$$\textcircled{a} S_w 0.6 = \frac{0.79 - 0.6}{0.6 - 0.502} = \frac{\partial f_w}{\partial S_w} = 1.93$$

$$\textcircled{a} S_w 0.7 \quad \frac{\partial f_w}{\partial S_w} = \frac{0.83 - 0.7}{0.6 - 0.443} = 0.83$$

$$\textcircled{a} S_w 0.8 \quad \frac{\partial f_w}{\partial S_w} = \frac{0.94 - 0.85}{0.65 - 0.343} = 0.293$$

S_{w2}	f_{w2}	$\frac{\partial f_w}{\partial S_w}$	Q_i	$\bar{S}_w$	E_D	N_p	W_{inr}	t, days	WOR_3	Q_0	Q
0.596	0.780	1.97	0.507	0.707	0.634	314,780	393,198	437	4.34	158	68
0.600	0.788	1.93	0.518	0.710	0.638	316,766	401,852	447	4.56	153	69
0.700	0.922	0.83	1.205	0.694	0.6175	306,588	934,814	1038	14.49	56	81
0.800	0.974	0.293	3.407	0.889	0.861	427,486	2,643,011	2,937	45.91	19	80

$\textcircled{a}$ S_{w2} of 0.80, we will have a WOR of 45.91

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- Calculate WORs exactly at breakthrough.

$$\text{WORs} = \frac{1.25}{1.02 \left(\frac{1}{0.78} - 1 \right)} = 4.34 \text{ stb/stb}$$

$$\text{WORs} = \frac{B_o}{B_w \left(\frac{1}{f_w} - 1 \right)}$$

$$\bar{S}_{w2} = S_{w2} + \frac{1 - f_{w2}}{\left(\frac{\partial f_w}{\partial S_w} \right)_{S_{w2}}} =$$

$$= 0.6 + \frac{(1 - 0.788)}{1.93}$$

$$= 0.710$$

~~WOR~~ =

$$Q_w = Q_o \text{ WORs.}$$

$$Q_o = \frac{i_w}{B_o + B_w \text{ WORs}}$$

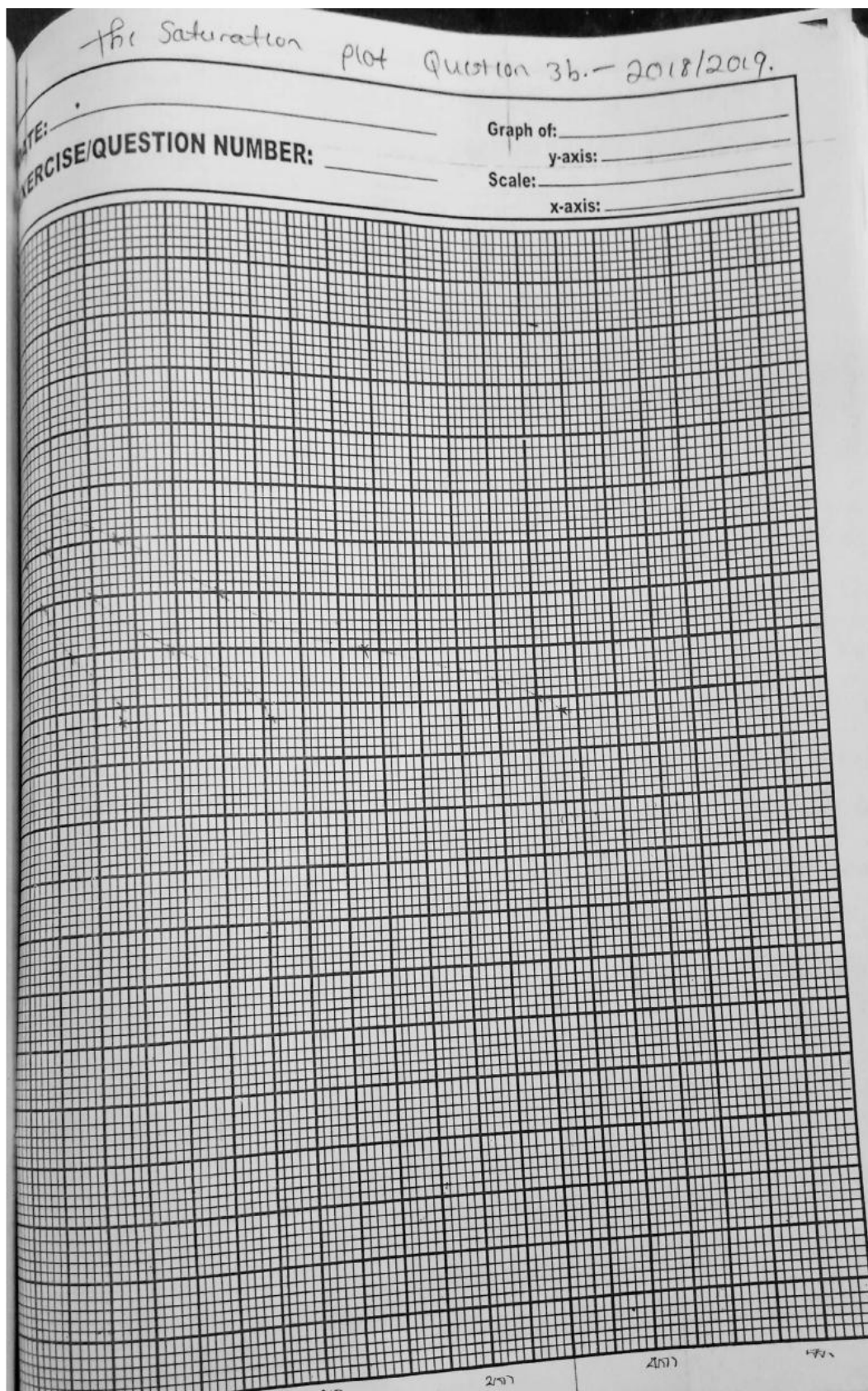
$$Q_o = \frac{900}{1.25 + 1.02 \times 4.34}$$

$$= 158.5 \text{ stb/d}$$

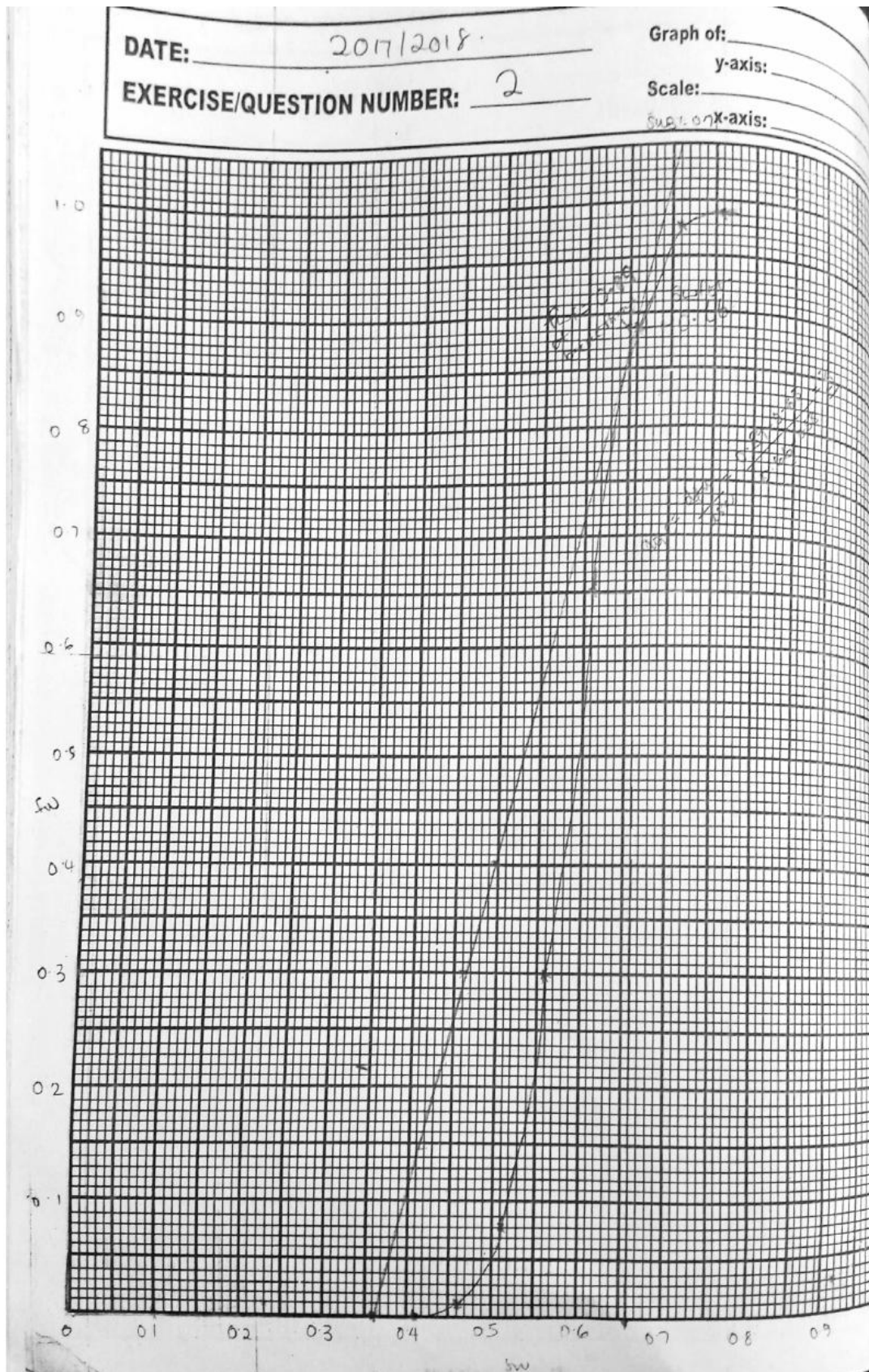
$$Q_w = 158 \times 4.34$$

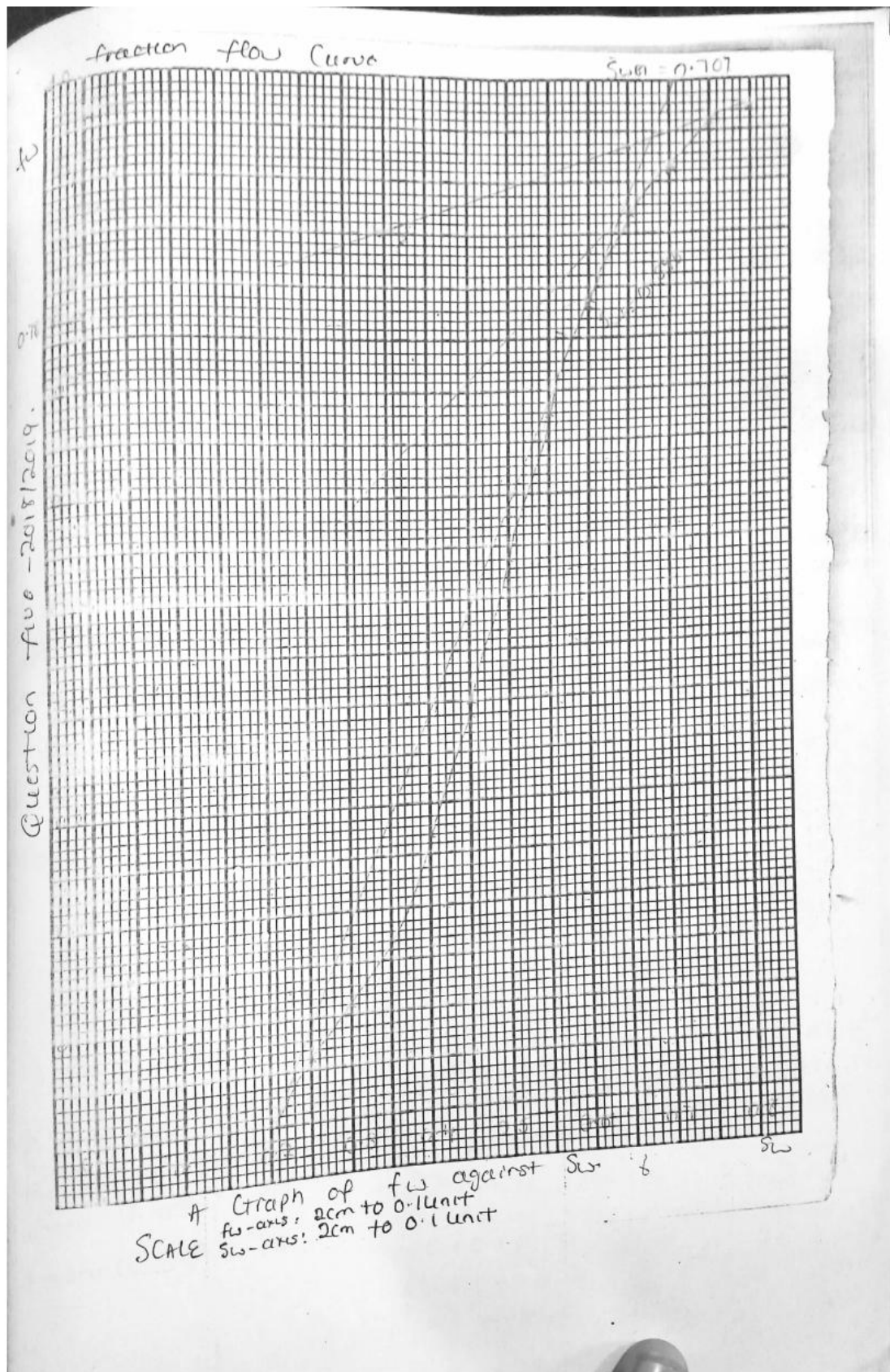
$$= \underline{685.72 \text{ stb/d}}$$

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QUESTION FIVE.

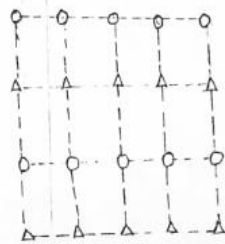
5b) Draw and Explain briefly the following regular injection patterns.

- ① Direct line drive.
- ② staggered line drive.
- ③ five SPOT
- ④ Seven SPOT
- ⑤ Nine SPOT

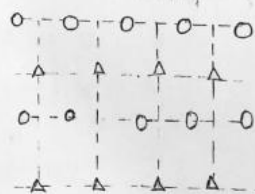
① Direct line drive: The lines of injection and production are directly opposed to each other. The pattern is characterized by two parameters:

a = distance between wells of the same type, and

b = distance between lines of injectors and producers.

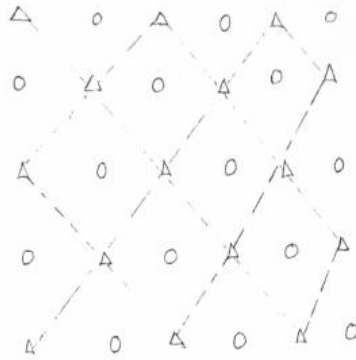


② staggered line drive: The wells are in lines as in the direct line, but the injectors and producers are no longer directly opposed but laterally displaced by a distance of $a/2$.

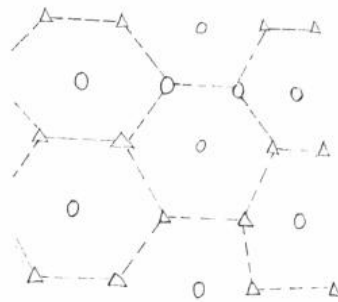


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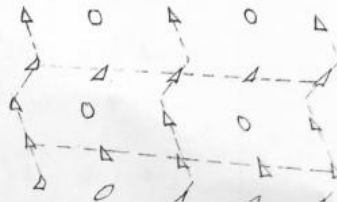
③ Five SPOT: This is a special case of the staggered, drive in which the distance between all like wells is constant, i.e. $a = 2d$. Any four injection wells thus form a square with a production well at the centre.



④ Seven SPOT: The injection wells are located at the corners of a hexagon with a production well at its center.



⑤ Nine Spot: This pattern is similar to that of the five spot but with an extra injection well drilled at the middle of each side of the square. The pattern essentially contains eight injectors surrounding one producer.



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